Experimental psychology
Lab Report

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Roll number: 30

M.Sc. Replica III

Session 2019-2021
Pigeon no. R-6

Submitted to

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Experimental Psychology

Experimental psychology can be defined as the scientific and empirical approach to the study of the mind. The experimental approach means that tests are administered to participants, with both control and experimental conditions.

This means that a group of participants are exposed to a stimulus (or stimuli), and their behavior in response is recorded. This behavior is compared to some kind of control condition, which could be either a neutral stimulus, the absence of a stimulus, or against a control group.

Experimental psychology is concerned with testing theories of human thoughts, feelings, actions, and beyond – any aspect of being human that involves the mind. This is a broad category that features many branches within it (e.g. behavioral psychology, cognitive psychology). Below, we will go through a brief history of experimental psychology, the aspects that characterize it, and outline research that has gone on to shape this field.

1.1 A Brief History of Experimental Psychology

As with anything, and perhaps particularly with scientific ideas, it's difficult to pinpoint the exact moment in which a thought or approach was conceived. One of the best candidates with which to credit the emergence of experimental psychology with is Gustav Fechner who came to prominence in the 1830's. After completing his Ph.D in biology at the University of Leipzig [1], and continuing his work as a professor, he made a significant breakthrough in the conception of mental states.

As Schultz and Schultz recount [2]: "An increase in the intensity of a stimulus, Fechner argued, does not produce a one-to-one increase in the intensity of the sensation … For example, adding the sound of one bell to that of an already ringing bell produces a greater increase in sensation than adding one bell to 10 others already ringing. Therefore, the effects of stimulus intensities are not absolute but are relative to the amount of sensation that already exists."

This ultimately meant that mental perception is responsive to the material world – the mind doesn't passively respond to a stimulus (if that was the case, there would be a linear relationship between the intensity of a stimulus and the actual perception of it), but is dynamically responsive to it. This conception ultimately shapes much of experimental psychology, and the grounding theory – that the response of the brain to the environment can be understood in calculable ways.

Fechner went on to research within this area for many subsequent years, testing new ideas regarding human perception. Over in Heidelberg, another German scientist, interested essentially in the problem of multitasking, began to detect and record his responses to different perceptual stimuli. The scientist was Wilhem Wundt, heavily influenced by the work of Gustav Fechner.

Wilhem Wundt is often credited with being "the father of experimental psychology" and is the founding point for many aspects of it. He began the first experimental psychology lab, scientific journal, and ultimately formalized the approach as a science. Wundt set in stone what Fechner had put on paper.

The next scientist to advance the field of experimental psychology was influenced directly by reading Fechner's book "Elements of Psychophysics". Hermann Ebbinghaus, also a German scientist, carried out the first properly formalized research into memory and forgetting, by using long lists of (mostly) nonsense syllables (such as: “VAW”, “TEL”, “BOC”) and recording how long it took for people to forget them.

Experiments using this list, concerning learning and memory, would take up much of Ebbinghaus’ career, and help cement experimental psychology as a science. There are many other scientists’ whose contributions helped pave the way for the direction, approach, and success of experimental psychology (Hermann von Helmholtz, Ernst Weber, and Mary Whiton Calkins, to name just a few), yet their work is beyond the scope of this post.

1.2 Definition

Defining any scientific field is in itself no exact science – there are inevitably aspects that will be missed. However, experimental psychology features at least three central components that define it: empiricism, falsifiability, and determinism. These features are central to experimental psychology but also many other fields within science.

Empiricism refers to the collection of data that can support or refute a theory. In opposition to purely theoretical reasoning, empiricism is concerned with observations that can be tested. It is based on the notion that all knowledge stems from sensory experience – that observations can be perceived and data surrounding them can be collected to form experiments.

Falsifiability is a foundational aspect of all contemporary scientific work. Karl Popper, a 20th century philosopher, formalized this concept – that for any theory to be scientific there must be a way to falsify it.

The Theory of Relativity is scientific, for example, because it is possible that evidence could emerge to disprove it. This means that it can be tested. An example of an unfalsifiable argument is that the earth is younger than it appears, but that it was created to appear older than it is – any evidence against this is dismissed within the argument itself, rendering it impossible to falsify, and therefore untestable.

Determinism refers to the notion that any event has a cause before it. Applied to mental states, this means that the brain responds to stimuli, and that these responses can ultimately be predicted, given the correct data.

These aspects of experimental psychology run throughout the research carried out within this field. There are thousands of articles featuring research that have been carried out within this vein – below we will go through just a few of the most influential and well-cited studies that have shaped this field, and look to the future of experimental psychology.

1.3 Scope

Scope of a field refers to the future a particular career holds, how it is applied, its value and importance in the society. Scope varies with culture, geography, technological advancements and some other factors. All in all, scope can be briefly defined as the pulls and pushes related to a field.

Experimental psychology is the most important branch of psychology. The credit for establishing psychology on a scientific basis goes to experimental method.

This method is now being used more and more in psychological studies.

The scope of Experimental Psychology is widening with the invention of new tools and instruments for experiments. Therefore, it is in the fitness of things that experimental psychology constitutes compulsory part of courses of psychology for the under-graduate and post-graduate students in universities everywhere in the world.

Experimental Psychology studies external behavior as well as the internal processes of the different stages of human development. Only those phenomenon fall outside its field which cannot be studied in controlled situations. The most important areas covered by experimental psychology include psycho-physics, animal psychology, learning psychology, psychology of individual differences, child psychology, educational psychology, clinical psychology and industrial psychology etc.

Due to the development of experimental psychology, other branches of psychology have managed to also develop their breadth of knowledge.

1.4 Implementation

The experimental method involves manipulating one variable to determine if changes in one variable cause changes in another variable. This method relies on controlled methods, random assignment and the manipulation of variables to test a hypothesis.

In order to understand how the experimental method works, there are some key terms you should first understand.

The independent variable is the treatment that the experimenter manipulates. This variable is assumed to cause some type of effect on another variable. If a researcher was investigating how sleep influences test scores, the amount of sleep an individual gets would be the independent variable.

The dependent variable is the effect that the experimenter is measuring. In our previous example, the test scores would be the dependent variable.

Operational definitions are necessary in order to perform an experiment. When we say something is an independent variable or dependent variable, we need to have a very clear and specific definition of the meaning and scope of that variable.

A hypothesis is a tentative statement or guesses about the possible relationship between two or more variables. In our earlier example, the researcher might hypothesize that people who get more sleep will perform better on a math test the next day. The purpose of the experiment is then to either support or fail to support this hypothesis.

1.5 Role

Experimental psychologists use scientific methods to collect data and perform research. Often, their work builds, one study at a time, to a larger finding or conclusion. Some researchers have devoted their entire career to answering one complex research question.

These psychologists work in a variety of settings, including universities, research centers, government agencies and private businesses. The focus of their research is as varied as the settings in which they work. Often, personal interest and educational background will influence the research questions they choose to explore.

In a sense, all psychologists can be considered experimental psychologists since research is the foundation of the discipline, and many psychologists split their professional focus among research, patient care, teaching or program administration. Experimental psychologists, however, often devote their full attention to research — its design, execution, analysis and dissemination.

Those focusing their careers specifically on experimental psychology contribute work across subfields. For example, they use scientific research to provide insights that improve teaching and learning, create safer workplaces and transportation systems, improve substance abuse treatment programs and promote healthy child development.

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Practical I

Muller lyer Illusion

Problem Statement

The purpose of this practical is to assess the molar lyer illusion of the subject.

Introduction

Here are a few basic things to look for and tactics to help ease the way.

1. **Perception**

In common terminology, perception is defined by Longman Dictionary of Contemporary English as "a) the way you think about something and your idea of what it is like; b) the way that you notice things with your senses of sight, hearing etc.; c) the natural ability to understand or notice things quickly." In philosophy, psychology, and cognitive science, perception is the process of attaining awareness or understanding of sensory information. The word "perception" comes from the Latin words *perceptio*, and means "receiving, collecting, action of taking possession, and apprehension with the mind or senses."

1.1 Perception Process

The perception process consists of three stages: selection, organization, and interpretation.

1.1.1 Selection

Selection is the first stage in the process of perception, during which we convert the environment stimuli into meaningful experience.

1.1.2 Organization

The second stage in perception process is organization. After selecting information from the outside world, we need to organize it in some way by finding certain meaningful patterns.

1.1.3 Interpretation

The third stage in perception is interpretation, which refers to the process of attaching meaning to the selected stimuli.

2. Sensation

Sensation is a process by which our sensory receptors and nervous system receive and represent stimulus energies from our environment.

2.1 Types of Sensations

Sensations are of three kinds:

2.1.1 Organic Sensations

Hunger, thirst, etc., are organic sensations. Organic sensations are produced by the conditions of the internal organs of the body.

2.1.2 Special Sensations

Sensations of colors, sounds, tastes, smells, pressures, heat, cold, etc., are special sensations. Special sensations are produced by the stimulation of the special sense-organs, viz., the eye, the ear, the tongue, the nose and the skin by special kinds of stimuli.

2.1.3 Motor Sensations

Sensations of movement are motor sensations. Motor sensations are produced by changes in the organs of movement, viz., muscles, tendons and joints.

3. Stimulus

A stimulus is anything that can trigger a physical or behavioral change. The plural of stimulus is stimuli. Stimuli can be external or internal.

4. Illusion

A misrepresentation of a "real" sensory stimulus—that is, an interpretation that contradicts objective "reality" as defined by general agreement. For example, a child who perceives tree branches at night as if they are goblins may be said to be having an illusion. An illusion is distinguished from a hallucination, an experience that seems to originate without an external source of stimulation. Both experiences is necessarily a sign of psychiatric disturbance, and both are regularly and consistently reported by virtually everyone.

4.1 Nature of Illusions

Illusions are special perceptual experiences in which information arising from "real" external stimuli leads to an incorrect perception, or false impression, of the object or event from which the stimulation comes.

Some of these false impressions may arise from factors beyond an individual's control (such as the characteristic behaviour of light waves that makes a pencil in a glass of water seem bent), from inadequate information (as under conditions of poor illumination), or from the functional and structural characteristics of the sensory apparatus (e.g., distortions in the shape of the lens in the eye). Such visual illusions are experienced by every sighted person.

Another group of illusions results from misinterpretations one makes of seemingly adequate sensory cues. In such illusions, sensory impressions seem to contradict the "facts of reality" or fail to report their "true" character. (For more-profound philosophical considerations, *see* epistemology.) In these instances the perceiver seems to be making an error in processing sensory information. The error appears to arise within the central nervous system (brain and spinal cord); this may result from competing sensory information, psychologically meaningful distorting influences, or previous expectations (mental set). Drivers who see their own headlights reflected in the window of a store, for example, may experience the illusion that another vehicle is coming toward them even though they know there is no road there.

4.2 Types of Illusions

4.2.1 Stimulus-distortion illusions

This type of illusory sense perception arises when the environment changes or warps the stimulus energy on the way to the person, who perceives it in its distorted pattern

4.2.2 Auditory Illusions

A common phenomenon is the auditory impression that a blowing automobile horn changes its pitch as it passes an observer on a highway. This is known as the Doppler effect, for Christian Doppler, an Austrian physicist, who in 1842 noted that the pitch of a bell or whistle on a passing railroad train is heard to drop when the train and the perceiver are moving away from each other and to grow higher when they are approaching each other. The sound heard is also affected by factors such as a wind blowing toward or away from the person.

Another auditory illusion was described in 1928 by Paul Thomas Young, an American psychologist, who tested the process of sound localization (the direction from which sound seems to come). He constructed a pseudophone, an instrument made of two ear trumpets, one leading from the right side of the head to the left ear and the other vice versa. This created the illusory impression of reversed localization of sound. While walking along the street wearing the pseudophone, he would hear footsteps to his right when they actually came from the left.

When two sources of sound in the same vicinity emit sound waves of slightly different frequencies (i.e., vibrations per second), there will come intervals when waves from both sources arrive at the ear in phase (simultaneously) and produce the experience of a combined, louder sound. These intervals of combined sound will be perceived as “beats,” or periodic alternations of sound intensity. When such auditory beats occur too rapidly to be discriminated, a harsh, continuous noise, commonly called interference, may result. Another case of interference results when two tones sounded together produce a subjectively heard third tone. When this third tone is lower in pitch than the original two, it is called a difference tone; i.e., its frequency is the difference between the frequencies of the two original tones. When the third tone is higher, it is called a summation tone; i.e., its frequency is the sum of the frequencies of the two original tones. Piano tuners depend in part upon their ability to hear these tones in a reliable way when tightening and loosening the strings in order to reach the correct pitch on the instrument.

4.2.3 Optical Illusions

Numerous optical illusions are produced by the refraction (bending) of light as it passes through one substance to another in which the speed of light is significantly different. A ray of light passing from one transparent medium (air) to another (water) is bent as it emerges. Thus, the pencil standing in water seems broken at the surface where the air and water meet; in the same way, a partially submerged log in the water of a swamp gives the illusion of being bent.

4.2.4 Perceiver-Distortion Illusions

Some illusions are related to characteristics of the perceiver, namely the functioning of the brain and the senses, rather than to physical phenomena that distort a stimulus. Many common visual illusions are perceptual: they result from the brain's processing of ambiguous or unusual visual information. Other illusions result from the aftereffects of sensory stimulation or from conflicting sensory information. Still others are associated with psychiatric causes.

4.2.5 Visual Perceptual Illusions

When an observer is confronted with a visual assortment of dots, the brain may group the dots that "belong together." These groupings are made on the basis of such things as observed similarity (e.g., red versus black dots), proximity, common direction of movement, perceptual set (the way one is expecting to see things grouped), and extrapolation (one's estimate of what will happen based on an extension of what is now happening).

Closure (a term used in Gestalt psychology) is the illusion of seeing an incomplete stimulus as though it were whole. Thus, one unconsciously tends to complete (close) a triangle or a square that has a gap in one of its sides. While a person watches a movie, closure occurs to fill the intervals between what are really rapidly projected still pictures—giving the illusion of uninterrupted motion.

4.2.6 Sensory Illusions

Many sensory illusions may be described as the aftereffects of the stimulation, or overstimulation, of the senses. Sensitivity in any of the senses may be measured as the just-perceptible intensity (threshold, or limen) of the appropriate stimulus.

4.2.7 Color Illusions

The normal human eye can detect about 130 gradations of colour in the visible spectrum (as in the rainbow), about 20 barely noticeable differences within a given colour, and about 500 variations of brightness. However, when two spots of equally bright light are observed in close succession, the first intensity may seem brighter. The first light may be said to serve the function of brightness adaptation (or adjustment) in the eye; therefore, the second light will fall on a partly adapted and therefore less sensitive retina. In a brief time, such excitement in the retina (or even in the brain) tends to subside, or fade. As a result of the fading traces of excitement, various hues of a given colour may appear to be lighter or darker when looked at successively.

4.2.8 Weight Illusions

The felt perception of differences in weights received experimental attention in 1899, when experiments indicated that a second weight feels either heavier or lighter than an immediately preceding identical weight. This illusion results partially from the expectancy of the person doing the lifting. Having lifted the first weight, the subject is “set” for a certain effort on the next try. If the second weight is lifted quickly and easily, it will feel lighter than the first; if it comes up more slowly, it will feel heavier. Expectancy, or set, is also often invoked in efforts to explain the size-weight illusion, in which a large cardboard box feels lighter than a smaller box even though both weigh the same.

4.2.9 Olfactory Illusions

Smell (olfactory) discrimination is influenced by any odour to which the olfactory structures already have adapted. Receptors in the nose, however, adapt quickly and cease to respond to a particular stimulus. This effect is called olfactory fatigue. Thus, an odour that is strong at first will gradually become imperceptible, as happens when one becomes unaware of the smell of one's own body. There also may be present the phenomenon of masking; this is a decrease in sensitivity to one odour after exposure to another (for example, a strong-smelling disinfectant).

4.2.10 Loudness Illusions

The human ear typically serves to distinguish between about 1,500 levels of pitch. For loudness, differential-threshold studies reveal about 325 separately perceived levels in the region of greatest auditory sensitivity (about 1,000 to 4,000 cycles per second). For humans, the number of discriminable tones is in the hundred thousands. Yet when two sounds are heard in close succession, the intensity or loudness of the second is judged by comparing it with the first. Thus, a murmur may sound loud when compared to a whisper, or a "deafening" noise may make all other sounds inaudible. The steady hum of an electric fan may help to diffuse the noises of traffic outside and thus improve the discrimination of sounds in the room.

4.2.11 Tactile Illusions

The skin contains numerous "spots" that respond selectively either to cold or to warmth but generally not to both. It can happen, however, that a very warm stimulus will produce a sensation of cold when placed on a spot that responds to cold. Thus, when a warm stimulus is perceived as cold, the illusion is called paradoxical cold.

Paradoxical heat, a less frequent experience, results from stimulating warm and cold spots simultaneously. It appears to be a fusion of warm and paradoxical cold effects, producing a strange, somewhat unpleasant sensation of “heat” that seems to be attended by uneasiness resembling that of pain. The sensation is sometimes called psychological heat.

Sudden temperature contrasts can play tricks on the tactile sense. If hot water is run over one hand and cold water over the other long enough for both to adjust to the temperatures and then both hands are plunged into lukewarm water, the cold hand will feel warm and the hot, cold. It would seem that, in plunging the cold-adapted hand, nerve cells for perceiving cold are suddenly inhibited and those for perceiving heat are suddenly stimulated, while in the heat-adapted hand the reverse takes place.

4.3 The Muller Lyer Illusion

The Muller Lyer illusion is based on the Gestalt principles of convergence and divergence: the lines at the sides seem to lead the eye either inward or outward to create a false impression of length.

The Muller-Lyer illusion is a well-known optical illusion in which two lines of the same length appear to be of different lengths. The illusion was first created by a German psychologist named Franz Carl Muller-Lyer in 1889.

First discovered in 1889 by F.C. Muller-Lyer, the illusion has become the subject of considerable interest and different theories have emerged to explain the phenomenon.

The Müller-Lyer Illusion is one among a number of illusions where a central aspect of a simple line image – e.g. the length, straightness, or parallelism of lines – appears distorted in virtue of other aspects of the image – e.g. other background/foreground lines, or other intersecting shapes. These are sometimes called 'geometrical-optical illusions' - and you can search for others in the Illusions Index.

Method

- **Participants**
  1. **Experimenter:** **Name:** Maida Maqbool **Age:** 21 **Gender:** F
  2. **Subject:** **Name:** Kainaat Yousaf **Age:** 21 **Gender:** F
- **Apparatus**

  Pencil, Ruler, Molar lyer card.

- **Procedure**

Experimenter gave very comfortable environment to the subject. Subject sat in a relax chair. There was white light in the lab. There was no one except the subject and experimenter. The subject was showing excitement to do this experiment. Then experimenter gave him all the instruction about experiment. The experimenter instructs the subject to draw the variable line out on the right side and then slowly bring it inwards to try to level the line. When the subjects is satisfied that it is equal to the line on the outside card, take the card from the subject and measure it and write it down. Thus the subject was given ten trials something like that ten trials from right to inside, ten trials from right to outside, ten trials from left to inside and ten trials from left to outside. The subject does everything and then experimenter thanked the subject and later the experimenter analysis the data.

Results

Table

Trials	**Right Hand Trial**		**Left Hand Trial**	
	Ascending	**Descending**	**Ascending**	**Descending**
1	3.0	6.3	3.9	6.2
2	2.9	6.8	3.6	6.3
3	3.0	6.2	3.7	6.0
4	3.5	6.3	3.8	6.4
5	3.4	6.8	3.0	6.4
6	3.4	6.6	3.6	6.2
7	3.0	6.5	3.6	6.5
8	3.3	6.3	3.7	6.6
9	3.2	6.5	3.2	6.3
10	3.9	6.3	3.6	6.4
Mean	M1 = 3.3	M2 = 6.7	M3 = 3.6	M4 = 6.3

Average of Ascending Trials

M1 = 3.3 M2 = 3.6

Point of subjective equality (PSE) = $\frac{\text{Right ascending + Left ascending}}{2}$

(PSE) = $\frac{3.3 + 3.6}{2}$ = 6.9

2 2

(PSE) = 3.5

Interval of uncertainty (IU) = 3.6 – 3.3 = **0.3**

Differential threshold (DL) = IU/2

DL = 0.3/2 = **0.15**

Average of descending trials

M2 = 6.7 M4 = 6.3

Point of subjective equality (PSE) = $\frac{\text{Right descending + Left descending}}{2}$

(PSE) = $\frac{6.7 + 6.3}{2} = \frac{13}{2}$

(PSE) = 6.5

Interval of uncertainty (IU) = 6.7 – 6.3 = **0.4**

Differential threshold (DL) = IU/2

DL = 0.4/2 = **0.2**

Conclusions

Result shows that the subject has illusion about arrows. There are a lot of illusions in some trials. And in some trials, there is very little illusion. Illusion is high in left-handed trials. The average of ascending trials of right hand and left hand is 3.3 and 3.6. Whereas, the average of descending trials of right hand and left hand is 6.7 and 6.3.

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Practical II

Word association

And

Reaction time

Problem Statement

The purpose of this practical is to assess the word association and reaction time of the subject.

Introduction

1. Learning

Psychologists often define learning as a relatively permanent change in behavior as a result of experience. The psychology of learning focuses on a range of topics related to how people learn and interact with their environments.

One of the first thinkers to study how learning influences behavior was psychologist "John B. Watson" who suggested that all behaviors are a result of the learning process. The school of thought that emerged from Watson's work was known as behaviorism. The behavioral school of thought proposed studying internal thoughts, memories, and other mental processes that were too subjective.

Psychology, the behaviorists believed, should be the scientific study of observable behavior. Behaviorism thrived during the first half of the twentieth century and contributed a great deal to our understanding of some important learning processes.

1.1 Types of Learning

The three major types of learning described by behavioral psychology are:

- Classical Conditioning
- Operant Conditioning
- Observational learning.

1.1.1 Classical Conditioning

Classical conditioning is a learning process in which an association is made between a previously neutral stimulus and a stimulus that naturally evokes a response.

For example, in Pavlov's classic experiment, the smell of food was the naturally occurring stimulus that was paired with the previously neutral ringing of the bell. Once an association had been made between the two, the sound of the bell alone could lead to a response.

1.1.2 Operant Conditioning

Operant conditioning is a learning process in which the probability of a response occurring is increased or decreased due to reinforcement or punishment. First studied by Edward Thorndike and later by B.F. Skinner, the underlying idea behind operant conditioning is that the consequences of our actions shape voluntary behavior.

Skinner described how reinforcement could lead to increases in behaviors where punishment would result in decreases. He also found that the timing of when reinforcements were delivered influenced how quickly a behavior was learned and how strong the response would be. The timing and rate of reinforcement are known as schedules of reinforcement.

1.1.3 Observational Conditioning

Observational learning is a process in which learning occurs through observing and imitating others. Albert Bandura's social learning theory suggests that in addition to learning through conditioning, people also learn through observing and imitating the actions of others.

As demonstrated in his classic "Bobo Doll" experiments, people will imitate the actions of others without direct reinforcement. Four important elements are essential for effective observational learning: attention, motor skills, motivation, and memory.

2. Law of Association

The Laws of Association explain how we learn and remember things. The philosopher Aristotle came up with the three basic Laws of Association: law of contiguity, law of similarity, and law of contrast.

The Law of Contiguity states that we associate things that occur close to each other in time or space. For example, if we think of thunder, we immediately think of lightning, since the two often occur one after the other.

The Law of Similarity states that when two things are very similar to each other, the thought of one will often trigger the thought of the other. For example, when we think of coffee, we often think of tea as well.

The Law of Contrast states that the thought of something is likely to trigger the thought of its direct opposite. For example, when we hear the word "hot," we often think of the word "cold."

These three basic laws were later revised and expanded by other associationists, but the Laws of Association can be considered as the seeds from which we can trace the beginnings of Psychology.

3. Reaction/Response Time

Reaction time or response time refers to the amount of time that takes places between when we perceive something to when we respond to it. It is the ability to detect, process, and respond to a stimulus.

Reaction time depends on various factors:

- **Perception**: Seeing, hearing, or feeling a stimulus with certainty is essential to having good reaction time. When the starter shoots the gun at the beginning of a race, the sound is received by the athlete's ears (they perceive the stimulus).
- **Processing**: In order to have good reaction time, it's necessary to be focused and understand the information well. Following the previous example, the runners, after hearing the gun, will be able to distinguish the sound from other background noise and know that it is time to start running (process the stimulus).
- **Response**: Motor agility is necessary in order to be able to act and have good response time. When the runners perceived and correctly processes the signal, they started moving their legs (respond to the stimulus).

Method

- **Participants**

  3. **Experimenter:** **Name:** Maida Maqbool **Age:** 21 **Gender:** F
  4. **Subject:** **Name:** Kainaat Yousaf **Age:** 21 **Gender:** F

- **Apparatus**

Pencil, Ruler, KENT-ROSANOFF test.

- **Procedure**

Experimenter gave very comfortable environment to the subject. Subject sat in a relax chair. There was yellow light in the lab. The subject was very excited. Then experimenter gave him all the instruction about experiment. Experimenter told the subject that he would say some words and as soon as any word came in mind said the word. Thus the experimenter said hundred words and the words that cameto the subject's mind were spoken. Then experimenter thanked the subject and later analysis the data.

Results

Table 1

Serial No.	Words	Response	Reaction Time (Sec)	Association
1	Table	Chair	2:18	Continuity
2	Dark	Night	2:42	Similarity
3	Music	Dance	2:75	Cause and effect
4	Sickness	Disease	3:00	Cause and effect
5	Man	Women	1:92	Contrast
6	Deep	Shallow	2	Similarity
7	Soft	Smooth	2:5	Similarity
8	Eating	Sugar	3	Continuity
9	Mountain	Beautiful	1:5	Continuity
10	House	Wife	2:3	Continuity
11	Black	White	2:5	Contrast
12	Mutton	Karahi	2:6	Similarity
13	Comfort	Winter	4:5	Continuity

14	Hand	Shake	3	Continuity
15	Short	Long	2:6	Contrast
16	Fruit	Vegetable	3	Contrast
17	Butterfly	Beautiful	3:5	Continuity
18	Smooth	Soft	2:4	Similarity
19	Command	Head	4:2	Cause and effect
20	Chair	Table	2:3	Continuity
21	Sweet	Heart	3	Continuity
22	Whistle	Watchman	2:6	Continuity
23	Women	Man	2	Contrast
24	Cold	Hot	2	Contrast
25	Slow	Fast	2:3	Contrast
26	Wish	Birthday	2	Continuity
27	River	Lake	2	Similarity
28	White	Black	1	Contrast
29	Beautiful	Butterfly	3:6	Continuity
30	Window	Door	2:7	Continuity
31	Rough	Register	2:8	Continuity
32	Citizen	Lahore	1	Continuity
33	Foot	Fingers	3	Similarity
34	Spider	Man	2	Continuity
35	Needle	Cloth	2:7	Continuity
36	Red	Blue	2	Similarity
37	Sleep	Awake	3	Contrast

38	Anger	Happiness	1:4	Contrast
39	Carpet	Red	6	Continuity
40	Girl	Boy	2	Contrast
41	High	Mountain	3	Similarity
42	Working	Women	1:7	Continuity
43	Sour	Sweet	1	Contrast
44	Earth	Quake	2:5	Continuity
45	Trouble	Shoot	6:7	Continuity
46	Soldier	Army	3:5	Similarity
47	Cabbage	Vegetable	1	Similarity
48	Hard	Soft	2	Contrast
49	Eagle	Mountain	3	Continuity
50	Stomach	Pain	2:7	Continuity
51	Stem	Leave	1:4	Similarity
52	Lamp	Light	2:6	Cause and effect
53	Dream	Sleep	3	Cause and effect
54	Yellow	White	1:5	Similarity
55	Bread	Egg	3	Continuity
56	Justice	Law	3	Cause and effect
57	Boy	Girl	2	Contrast
58	Light	Lamp	4	Cause and effect
59	Health	Wealth	2	Continuity
60	Bible	Book	5:5	Continuity
61	Memory	Lose	2:5	Continuity

62	Sheep	Black	2:6	Continuity
63	Bath	Soap	4	Continuity
64	Cottage	House	3:3	Similarity
65	Swift	Slow	1:7	Contrast
66	Blue	Red	1	Similarity
67	Hungary	Game	3:7	Continuity
68	Priest	Man	2:5	Similarity
69	Ocean	Lake	3	Similarity
70	Head	Foot	2	Contrast
71	Stove	Fire	1:2	Cause and effect
72	Long	Short	1:7	Contrast
73	Religion	Islam	2	Continuity
74	Whiskey	Haraam	3	Continuity
75	Child	Hood	2	Continuity
76	Bitter	Butter	2:8	Continuity
77	Hammer	Nail	2:7	Continuity
78	Thirsty	Crow	2	Continuity
79	City	Lahore	3	Continuity
80	Square	Rectangle	2	Similarity
81	Butter	Sweet	2:7	Similarity
82	Doctor	Psychologist	1	Similarity
83	Loud	Light	6:5	Contrast
84	Thief	Jewelry	5:7	Continuity
85	Lion	Lioness	1:4	Contrast

86	Joy	Land	3:3	Continuity
87	Bed	Rest	2:5	Continuity
88	Heavy	Light	3	Contrast
89	Tobacco	Cancer	3:2	Cause and effect
90	Baby	Cute	3	Continuity
91	Moon	Light	2	Cause and effect
92	Scissors	Paper	2	Continuity
93	Quiet	Loud	3	Contrast
94	Green	Yellow	2	Similarity
95	Salt	Sugar	2	Contrast
96	Street	Light	1:9	Continuity
97	King	Queen	2:4	Contrast
98	Cheese	Pizza	1:9	Continuity
99	Blossom	Bubble	1	Continuity
100	Afraid	Scare	6	Continuity

Average reaction time of the subject

$$= \frac{\text{Sum of reaction time}}{\text{Total numbers}}$$

$$= \frac{264.2}{100}$$

= 2.642

Table 2

Serial No.	Type of Association	No. of Response	Total Time(Sec.)
1	Continuity	45	131.03
2	Contrast	23	56.9
3	Similarity	22	47.32
4	Cause and Effect	10	28.95

Conclusions

Result shows that there is a lot of continuity response of the subject. And the average reaction time is 2.6. In some responses the subject took one to two seconds but in some responses the subject took five and six seconds like in response of the word "Trouble" the subject took six seconds. And in response of the word "Bible" the subject took five seconds. Whereas there is small numbers of response of "similarity", "contrast" and "cause and effect".

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Practical III

Personality types, Free association

And

Reaction time

Problem Statement

The purpose of this practical is to assess the personality trait and free word association along with reaction time of the subject.

Introduction

1. **Personality**

A characteristic way of thinking, feeling, and behaving. Personality embraces moods, attitudes, and opinions and is most clearly expressed in interactions with other people. It includes behavioral characteristics, both inherent and acquired, that distinguish one person from another and that can be observed in people's relations to the environment and to the social group. (Holzman, 2020).

The term personality has been defined in many ways, but as a psychological concept two main meanings have evolved. The first pertains to the consistent differences that exist between people: in this sense, the study of personality focuses on classifying and explaining relatively stable human psychological characteristics. The second meaning emphasizes those qualities that make all people alike and that distinguish psychological man from other species; it directs the personality theorist to search for those regularities among all people that define the nature of man as well as the factors that influence the course of lives. This duality may help explain the two directions that personality studies have taken: on the one hand, the study of ever more specific qualities in people, and, on the other, the search for the organized totality of psychological functions that emphasizes the interplay between organic and psychological events within people and those social and biological events that surround them. The dual definition of personality is interwoven in most of the topics discussed below. It should be emphasized, however, that no definition of personality has found universal acceptance within the field.

The study of personality can be said to have its origins in the fundamental idea that people are distinguished by their characteristic individual patterns of behaviour—the distinctive ways in which they walk, talk, furnish their living quarters, or express their urges. Whatever the behavior, personologists—as those who systematically study personality are called—examine how people differ in the ways they express themselves and attempt to determine the causes of these differences. Although other fields of psychology examine many of the same functions and processes, such as attention, thinking, or motivation, the personologist places emphasis on how these different processes fit together and become integrated so as to give each person a distinctive identity, or personality. The systematic psychological study of personality has emerged from a number of different sources, including psychiatric case studies that focused on lives in distress, from philosophy, which explores the nature of man, and from physiology, anthropology, and social psychology.

2. The Five Factor Theory

Costa (born in 1942) and McCrae (born in 1949) followed in the footsteps of Eysenck, but they expanded slightly upon the number of second order factors. The result of their efforts became one the most widely respected perspectives on personality structure today: the **Five-Factor Model** of personality. Indeed, the Five-Factor Model has been so well researched, research that has supported and expanded the original conception, that Costa and McCrae believe it now deserves to be referred to as the **Five-Factor Theory** .

Paul T. Costa, and Robert R. McCrae are an extraordinarily productive research team that has worked together since they first met in Boston in 1975. Their more than 250 publications on personality traits and the Five Factor model have had a profound effect on personality assessment, theory, and research.

2.1 The Five-Factor Model

The five-factor model organizes all personality traits along a continuum of five

factors: openness, extraversion, conscientiousness, agreeableness, and neuroticism. ▸ Many psychologists believe that the total number of personality traits can be reduced to five factors, with all other personality traits fitting within these five factors. According to this model, a **factor** is a larger category that encompasses many smaller personality **traits**. The five factor model was reached independently by several different psychologists over a number of years.

2.2 HISTORY

The Big Five was originally derived in 1970's by two independent research teams – Paul Costa and Robert McCrae

These five dimensions were derived by asking thousands of people hundreds of questions and then analyzing the data with the statistical procedure known as **factor analysis**

The Big Five is now the most widely accepted and used model of personality

The five factor theory is now amongst the newest models developed for the description of personality and this model shows promise to be amongst the most practical and applicable models available in the field of personality psychology

Because this model was developed independently by different theorists, the names of each of the five factors—and what each factor measures—differ according to which theorist is referencing it. Paul Costa's and Robert McCrae's version, however, is the most well-known today and the one called to mind by most psychologists when discussing the five factor model. The acronym OCEAN is often used to recall Costa's and McCrae's five factors, or the Big Five personality traits: Openness to Experience, Conscientiousness, Extraversion, Agreeableness, and Neuroticism

2.3 The Big Five

Psychologists in today's society believe that personality can be divided into five basic dimensions. They are referred to as 'Big 5'personality traits. The idea was studied by Costa and McCrae

2.3.1 Definitions of the big 5 traits

- **Neuroticism:** A tendency to easily experience unpleasant emotions such as anxiety, anger or depression
- **Extroversion:** Energy, and the tendency to stimulation and the company of others.
- **Agreeableness:** A tendency to be compassionate and cooperative rather than suspicious and antagonistic towards others.
- **Conscientiousness:** A tendency to show self-discipline, act dutifully and aim for achievement.
- **Openness to experience:** Appreciation for art, emotion, adventure, and unusual ideas; imaginative and curious.

2.3.2 The Big Five Across Cultures

In order to evaluate the cross-cultural application of the FiveFactor Model, Robert McCrae has suggested that we need to address the issue in three ways. **Transcultural** analyses look for personality factors that transcend culture. In other words, personality factors that are universal, or common to all people. **Intracultural** analyses look at the specific expression of traits within a culture. And finally, **intercultural** analyses compare trait characteristics between cultures (see Allik & McCrae, 2002). In 2002, McCrae and Allik published The Five-Factor Model of Personality Across Cultures, a collection of research in which a variety of investigators examined the applicability of the Five-Factor Model (FFM) in a wide variety of cultures

In proposing a Five-Factor Theory of personality, McCrae and Costa addressed the nature of personality theories themselves:

A theory of personality is a way of accounting for what people are like and how they act; a good theory explains a wide range of observations and points researchers in the right direction for future research. Freudian theory pointed researchers toward the study of dreams, but decades of research have yielded very little by way of supportive evidence…Trait theory pointed researchers toward general styles of thinking, feeling, and acting, and has resulted in thousands of interesting and useful findings. That is why most personality psychologists today prefer trait theory to psychoanalysis…But…there is more to human personality than traits. (pp 184-185; McCrae & Costa, 2003)

They propose that there are three central components to personality: **basic tendencies** (which are the five personality factors), **characteristic adaptations**, and **self-concept** (a highly adapted and extensively studied form of characteristic adaptation). The basic tendencies interact with three peripheral components that mark the interface with systems outside personality. There are the biological inputs to the basic tendencies, the external environment, and **objective biography** (all that a person does and experiences). Connecting all of these components are **dynamic processes**, such as perception, coping, role playing, reasoning, etc. Although this theory is newer, it does account for one of the most important issues challenging trait theories in general: how does one account for the general consistency of traits, yet the potential for, and occasional observation of, change in personality? Simply, the basic tendencies are consistent, whereas the characteristic adaptations are subject to change, both as a result of dramatic environmental influences and due to changes associated with aging (McCrae & Costa, 2003).

2.4 Neo Personality Inventory

The Revised NEO Personality Inventory (NEO PI-R) is a personality inventory that examines a person's Big Five personality traits (openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism). In addition, the NEO PI-R also reports on six subcategories of each Big Five personality trait (called facets).

Historically, development of the Revised NEO PI-R began in 1978 with the publication of a personality inventory by Costa and McCrae. These researchers published three updated versions of their personality inventory in 1985, 1992, and 2005 which are called the **NEO PI, NEO PI-R** (or Revised NEO PI), and NEO PI-3, respectively. The revised inventories feature updated norms

Method

- **Participants**

5. **Experimenter:** **Name:** Maida Maqbool **Age:** 21 **Gender:** F
6. **Subject:** **Name:** Kainaat Yousaf **Age:** 21 **Gender:** F

- **Apparatus**

Pencil, Ruler, NEO card.

- **Procedure**

Experimenter gave very comfortable environment to the subject. Subject sat in a relax chair. There was white light in the lab. The subject was showing excitement to do this experiment. Then experimenter gave him all the instruction about experiment. Experimenter gave the subject test and subject began to do it. The subject answers all questions and then experimenter thanked the subject and later the experimenter analysis the data.

Results

The personality type of the subject was "Neuroticism"

Table

Serial no.	Associations	Frequency	Average Reaction Time
1	Continuity	45	2.79
2	Contrast	23	2.47
3	Similarity	22	2.366
4	Cause and Effect	10	2.90

Conclusions

After analyzed the results, experimenter conclude that the personality type of the subject is low on neuroticism. But other personality types were very low. This means that the dominant personality type is neuroticism, which shows that subject is emotionally stable, happier and more satisfied with life because the level of neuroticism was low.

Association of continuity of the subject was high in frequency. Whereas, the other types of association were low in frequency. Average reaction time of the association of continuity was also more.

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Practical IV

Absolute threshold

And

Differentiate threshold

Problem Statement

The purpose of this practical is to assess the absolute threshold and differentiate threshold of the subject sense of taste.

Introduction

1. **Absolute threshold**

The minimal amount of energy necessary to stimulate the sensory receptors.

The method of testing for the absolute threshold is similar for different sensory systems. Thus, the tester can briefly present a light or a sound (or any other kind of stimulus) at different, low intensities until the observer is unable to detect the presence of the stimulus. In such a task, the person may undergo thousands of trials before the researcher can determine the threshold.

While the absolute threshold is a useful concept, it does not exist in reality. That is, on one occasion, an individual might be unable to detect a certain faint light.

1.1 Examples of Absolute Threshold

Our senses are examples of absolute threshold.

1.1.1 Vision

The amount of light present if someone held up a single candle 30 mi (48 km) away from us, if our eyes were used to the dark. If a person in front of you held up a candle and began backing up at the rate of one foot (30 cm) per second, that person would have to back up for 44 hours before the flame became invisible.

1.1.2 Hearing

The ticking of a watch in a quiet environment at 20 ft (6 m).

1.1.3 Taste

One drop on quinine sulfate (a bitter substance) in 250 gal (946 l) of water. Quinine is one of the components of tonic water.

1.1.4 Smell

One drop of perfume in a six-room house. This value will change depending on the type of sub-stance we smell.

1.1.5 Touch

The force exerted by dropping the wing of a bee onto your cheek from a distance of one centimeter (0.5 in). This value will vary considerably depending on the part of the body involved.

But on a subsequent occasion, may detect it. In addition, scientists cannot determine with absolute certainty how much energy is present in a light because of limits to the physics of measurement. As a result, psychologists often define the threshold as the lowest intensity that a person can detect 50 percent of the time.

A number of different factors can influence the absolute threshold, including the observer's motivations and expectations, and whether the person is adapted to the stimulus. Scientists have discovered that cognitive processes can influence the measurement of the threshold and that it is not as simple as once understood. Psychologists have also studied how different two stimuli have to be in order to be noticed as not being the same. Such an approach involves what are called difference thresholds.

2. Difference Threshold

"The minimum change in intensity required to produce a detectable change in sensory experience (this is also known as a Just Noticeable Difference or JND)".

The just noticeable difference (JND), also known as the difference threshold, is the minimum level of stimulation that a person can detect 50 percent of the time. For example, if you were asked to hold two objects of different weights, the just noticeable difference would be the minimum weight difference between the two that you could sense half of the time.

It is important not to confuse the just noticeable difference and the absolute threshold. While the difference threshold involves the ability to detect differences in stimulation levels, the absolute threshold refers to the smallest detectable level of stimulation.

The absolute threshold for sound, for example, would be the *lowest volume level* that a person could detect. The just noticeable difference would be the smallest *change in volume* that a person could sense.

2.1 Development of the Concept

The difference threshold was first described by a physiologist and experimental psychologist named Ernst Weber and later expanded upon by psychologist Gustav Fechner.

Weber's Law, also sometimes known as the Weber-Fechner Law, suggests that the just noticeable difference is a constant proportion of the original stimulus.

The just noticeable difference applies to a wide variety of senses including touch, taste, smell, hearing, and sight. It can apply to things such as brightness, sweetness, weight, pressure, and noisiness, among other things.

Method

- **Participants**

1. **Experimenter:** **Name:** Maida Maqbool **Age:** 21 **Gender:** F
2. **Subject:**

Subject	Name	Age	Gender
1	Kainaat yousaf	21	F
2	Fareeha Kousar	21	F
3	Tahreem	21	F
4	Fakiha Rafi	22	F
5	Tahira	21	F

- **Apparatus**

Pencil, Ruler, Glasses, Sugar, Water, Spoon

- **Procedure**

Experimenter gave very comfortable environment to the subjects. Subjects were sit in a relax chairs There was white light in the lab. The subjects were very excited because this was a group experiment. Then experimenter gave them all the instruction about experiment. The experimenter told the subjects to pick up a glass of water one by one and drink it and tell me where you feel more sugar. The subjects followed the experimenter's instruction and then experimenter thanked the subjects and later the experimenter analysis the data.

Results

Table

Subject	Absolute Threshold	2nd Threshold Point	Differential Threshold
1	0.5	0.85	0.35
2	0.35	1.35	1
3	0.5	0.9	0.35
4	0.6	0.95	0.35
5	0.6	1	1

Conclusion

Results show that every subject has its own threshold level of sense of taste. Subject no. 1 level of absolute threshold is 0.5. And the second point of threshold where the subject feels sugar again is 0.85. Exactly the same, the other subjects have their own sense of taste.

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Practical V

Intentional and Unintentional Learning

And

Memory processing

Problem Statement

The purpose of this practical is to assess intentional and unintentional learning and memory processing of the subject and measuring memory.

Introduction

1. **Memory**

"Memory is the process of maintaining information over time." (Matlin, 2005)

"Memory is the means by which we draw on our past experiences in order to use this information in the present' (Sternberg, 1999).

Memory is the ability to take in information, encode it, store it, and retrieve it at a later time.

1.1 Background

Already in the 19th century, the recognition that the number of neurons in the brain doesn't increase significantly after reaching adulthood suggested to early neuroanatomists that memories aren't primarily stored through the creation of neurons, but rather through the strengthening of connections *between* neurons (Ramón y Cajal, 1894). In 1966, the breakthrough discovery of long-term potentiation (LTP) suggested that memories may be encoded in the strength of synaptic signals between neurons (Bliss and Lømo, 1973). And so we started understanding memory as a neuro-chemical process. The studies by Eric Kandel of the *Aplysia californica*, for which he won the Nobel prize, for example, show that classical conditioning is a basic form of memory storage and is observable on a molecular level within simple organisms (Kandel et al., 2012). This in effect expanded the definition of memory to include storage of information in the neural networks of simple lifeforms. Increasingly, researchers are exploring the chemistry behind memory development and recall, suggesting these molecular processes can lead to psychological adaptations (e.g., Coderre et al., 2003; Laferrière et al., 2011).

Memory is today defined in psychology as the faculty of encoding, storing, and retrieving information (Squire, 2009). Psychologists have found that memory includes three important categories: sensory, short-term, and long-term. Each of these kinds of memory have different attributes, for example, sensory memory is not consciously controlled, short-term memory can only hold limited information, and long-term memory can store an indefinite amount of information.

Key to the emerging science of memory is the question of how memory is consolidated and processed. Long-term storage of memories happens on a synaptic level in most organisms (Bramham and Messaoudi, 2005), but, in complex organisms like ourselves, there is also a second form of memory consolidation: *systems consolidation* moves, processes, and more permanently stores memories (Frankland and Bontempi, 2005). Today, there are many models of how memory is consolidated in cognition. Single-system models posit that the hippocampus supports the neocortex in encoding and storing long-term memories through strengthening connections, finally leading the memory to become independent from the hippocampus (Ibid.). Multiple-trace theory instead proposes that each memory has a unique code or memory trace, which continues to involve the hippocampus to an extent (Hintzman and Block, 1971; Hintzman, 1986, 1990; Whittlesea, 1987; Versace et al., 2014; Briglia et al., 2018). In another theory, memory is understood as a form of negative entropy or rich energy (Wiener, 1961, 1988), which is then processed in a way that minimizes the expenditure of energy by the brain (Friston, 2010; Van der Helm, 2016). Our heightened capacity to store information may be due to our ability to reduce disorder and process large amounts of information rapidly, a necessarily non-linear process (Wiener, 1961, 1988). The forgetting and fading of memories is also understood as being an important aspect of the functioning and utility of these memories (Staniloiu and Markowitsch, 2012). As with a computer hard drive, memories can also be "corrupted" – false memories are commonly studied within forensic psychology (Loftus, 2005). Together, these advances highlight how different kinds of memory storage are non-linear – that is, subject to complex systems interactions – contextual, and plastic. They also shed light on why, and how, we are able to live with such large quantities of information. It may not be that Funes has the special ability to remember everything, but that he *lacks* our ability to incorporate, and sort through, a potentially infinite amount of information.

The advance of the fields of genetics and epigenetics has also given us new metaphors to describe memory. We understand DNA as a structure that carries information that we call "genetic code" – kind of like a computer chip for biological processes. Today, the metaphor has come full circle and we can now use DNA to store and extract digital data (Church et al., 2012). The study of epigenetics suggests that simple lifeforms pass on memories across generations through genetic code (Klosin et al., 2017; Posner et al., 2019), suggesting a need to study whether humans and other complex life forms may do so as well. With these advances, our understanding of how memory is stored has expanded once again.

Further, we can now store memory in places that we haven't been able to before. Smartphones, mind-controlled prosthetic limbs, and Google Glasses all offer new ways to store information and thereby interact with our surroundings. Our ability to produce information alters how we perceive the world, with far-reaching implications. As Stephen Hawking, the Nobel prize-winning physicist explained in his 1996 lecture, “Life in the universe,”

What distinguishes us from [our ancestors], is the knowledge that we have accumulated over the last 10000 years, and particularly, over the last three hundred. I think it is legitimate to take a broader view, and include externally transmitted information, as well as DNA, in the evolution of the human race (Hawking, 1996).

The sheer quantity of available information today, as well as developments in an understanding of memory – from fixed and physical to dynamic, chemical, and a process of rich energy transfer – lead to a very different picture of memory than the one we had 100 years ago. Memory seems to exist everywhere, from an *Aplysia*’s ganglion to DNA to a hard drive.

To account for these developments, cognitive scientists now propose that human cognition is actually extended beyond the brain in ways that theories of the mind did not previously recognize (Clark and Chalmers, 1998; Clark, 2008). This approach is being called 4E cognition (Embodied, Embedded, Extended, and Enactive). For example, enactivism posits that cognition is a dynamic interaction between an organism and its environment (Varela et al., 1991; Chemero, 2009; Menary, 2010; Rowlands, 2010; Favela and Chemero, 2016; Briglia et al., 2018). According to this framework, cognition is a process of incorporation between the environment and the body/brain/mind. To be clear, cognition is not incorporated *in* the surroundings, only the *corpus* can incorporate, and thus cognition (or what we call “mind”) is a product of the interaction between the brain, the body, and the environment.

1.2 Stages of memory

There are three stages of memory.

1.2.1 Encoding

The process of receiving, processing, and combining information. When information comes into our memory system (from sensory input), it needs to be changed into a form that the system can cope with, so that it can be stored.

There are three main ways in which information can be encoded (changed): Visual (picture), Acoustic (sound), Semantic (meaning).

1.2.2 Storage

The creation of a permanent record of the encoded information. Storage is the second memory stage or process in which we maintain information over periods of time. This concerns the nature of memory stores, i.e., where the information is stored, how long the memory lasts for (duration), how much can be stored at any time (capacity) and what kind of information is held.

1.2.3 Retrieval

The calling back of stored information in response to some cue for use in a process or activity. The third process is the retrieval of information that we have stored. We must locate it and return it to our consciousness. Some retrieval attempts may be effortless due to the type of information. This refers to getting information out storage. If we can’t remember something, it may be because we are unable to retrieve it.

1.3 Types of Memory

There are three types of personality.

1.3.1 Sensory Memory

Sensory memory allows individuals to retain impressions of sensory information after the original stimulus has ceased. One of the most common examples of sensory memory is fast-moving lights in darkness. Two other types of sensory memory have been extensively studied: echoic memory (the auditory sensory store) and haptic memory (the tactile sensory store).

1.3.2 Short Term Memory

Short-term memory (STM) is the second stage of the multi-store memory model proposed by the Atkinson-Shiffrin. The duration of STM seems to be between 15 and 30 seconds, and the capacity about 7 items. Short-term memory is also known as *working memory*. It holds only a few items (research shows a range of 7 +/- 2 items) and only lasts for about 20 seconds. However, items can be moved from short-term memory to long-term memory via processes like *rehearsal*.

1.3.3 Long Term Memory

Long-term memory (LTM) the final stage of the multi-store memory model proposed by the Atkinson-Shiffrin, providing the lasting retention of information and skills. Long-term memories are all the memories we hold for periods of time longer than a few seconds; long-term memory encompasses everything from what we learned in first grade to our old addresses to what we wore to work yesterday. Long-term memory has an incredibly vast storage capacity, and some memories can last from the time they are created until we die.

4. Learning

Psychologists often define learning as a relatively permanent change in behavior as a result of experience. The psychology of learning focuses on a range of topics related to how people learn and interact with their environments.

Method

- **Participants**

1. **Experimenter:** **Name:** Maida Maqbool **Age:** 21 **Gender:** F
2. **Subject:** **Name:** Kainaat Yousaf **Age:** 21 **Gender:** F

- **Apparatus**

Pencil, Ruler, 16 neutral pictures.

- **Procedure**

Experimenter gave very comfortable environment to the subject. Subject sat in a relax chair. There was white light in the lab. There was no one except the subject and experimenter. The subject was showing excitement to do this experiment. The experimenter did not give any instruction to the subject and started showing some pictures for ten seconds. After showing the pictures, the experimenter asked the subject about the pictures. The experimenter then instructs the subject about the experiment. Then the experimenter showed some pictures to the subject again. Then experimenter asked again about these pictures. After completing the experiment, experimenter thanked the subject and later analysis the data.

Results

Table

Group 1			Group 2		
Picture No.	Description	Percentage	Picture No.	Description	Percentage

1	Full	90%	1	Full	80%
2	Full	90%	2	Full	85%
3	Partial	50%	3	Full	90%
4	Partial	60%	4	Partial	60%
5	No	5%	5	Full	90%
6	No	2%	6	Partial	65%

Conclusion

Results show that intentionally learning is better than unintentionally learning. As the above table shows that when the subject was not know about the experiment, the subject could not remember those pictures. But when the experimenter told the subject about the experiment, the subjects remembered the pictures and explain very well.

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Pigeon Experiment

R-6

Table of contents

Introduction to Learning

Psychologists often define learning as a relatively permanent change in behavior as a result of experience. The psychology of learning focuses on a range of topics related to how people learn and interact with their environments.

One of the first thinkers to study how learning influences behavior was psychologist “John B. Watson” who suggested that all behaviors are a result of the learning process. The school of thought that emerged from Watson's work was known as behaviorism. The behavioral school of thought proposed studying internal thoughts, memories, and other mental processes that were too subjective.

Psychology, the behaviorists believed, should be the scientific study of observable behavior. Behaviorism thrived during the first half of the twentieth century and contributed a great deal to our understanding of some important learning processes.

Types of Learning

The three major types of learning described by behavioral psychology are:

- Classical Conditioning
- Operant Conditioning
- Observational learning.

Classical Conditioning

Classical conditioning is a learning process in which an association is made between a previously neutral stimulus and a stimulus that naturally evokes a response.

For example, in Pavlov's classic experiment, the smell of food was the naturally occurring stimulus that was paired with the previously neutral ringing of the bell. Once an association had been made between the two, the sound of the bell alone could lead to a response.

Operant Conditioning

Operant conditioning is a learning process in which the probability of a response occurring is increased or decreased due to reinforcement or punishment. First studied by Edward Thorndike and later by B.F. Skinner, the underlying idea behind operant conditioning is that the consequences of our actions shape voluntary behavior.

Skinner described how reinforcement could lead to increases in behaviors where punishment would result in decreases. He also found that the timing of when reinforcements were delivered influenced how quickly a behavior was learned and how strong the response would be. The timing and rate of reinforcement are known as schedules of reinforcement.

Observational Conditioning

Observational learning is a process in which learning occurs through observing and imitating others. Albert Bandura's social learning theory suggests that in addition to learning through conditioning, people also learn through observing and imitating the actions of others.

As demonstrated in his classic "Bobo Doll" experiments, people will imitate the actions of others without direct reinforcement. Four important elements are essential for effective observational learning: attention, motor skills, motivation, and memory.

Preparation for Pigeon Experiments

Purchase of pigeon

Pigeon was purchased from a reputable breeder. While purchasing the pigeon, some precautions were kept in mind including:

Pigeon's age should be at least two years as the weight of younger bird continually increases and it cannot provide a stable baseline for food deficiency. Pigeon should be sound, vigorous and energetic. Pigeon should have clean feet because the feathered can become layered with droppings. Pigeon has no congenital behavior or structure that could interfere with establishing of strong pecking operant. The pigeon which satisfy the maximum requirements of the experiment was purchased. It was a male pigeon and white and black color

General Care and Housing

The room in which pigeon was kept was dry, well ventilated and had the light as much as possible. The ventilator fan was available for to help to reduce odors. Grain was kept in the covered and metal containers that keep the food dry, clean and away from rats. Food and water container were separated. Fresh and clean water was provided to the pigeon. Water container was adjusted in outside the cage where pigeon was not able to walk over it. Pigeons are tough dry and healthy animal and diseases can be prevented if proper ventilation, clean water and proper food are given. Even under optimal conditioned, and an occasional bird may become sick. The birds were isolated immediately and the cage disinfected with Formalin solution. The home cage was identified with the names of experimenters and the name, number, and weight of the bird

Handling

In catching, holding or carrying a pigeon it is most important to support the pigeon and keep it wings folded. The pigeon was handled carefully. The pigeon was hold firmly and gently. The distractions such as noises, movements were avoided when holding the pigeon, because it affects the pigeon behavior emotionally. While holding the bird, its head was facing the experimenter and his tail away from experimenter. Pigeon was supported in palm of hands, holding down its wings with thumbs and securing his legs between first and second fingers. The pigeon was hold strongly when he tries to struggle. Care should be taken while transferring the pigeon from one hand to another.

Calculation of Ad libitum Weight

For the first few days after the arrival of the pigeon in the laboratory, the Pigeon had food water and grain all the times. He weighted at approximately the same time, everyday and record was maintained about the pigeon weights. The pigeon gained weight during the first days, because he was on free food. But soon his weight becomes stabilized. He had the constant weight that was called his ad libitum weight. By taking average of the last three days weight we get the ad-lib weight then are 272.3 grams.

Formula

$$\text{Adlib Weight} = \frac{D1 + D2 + D3}{3}$$

$$= \frac{272 + 273 + 272}{3}$$

$$= 272.3$$

Calculation of Experimental Weight

When the ad lib weight was determined, then the pigeon weight was reduced approximately 85% gradually. The pigeon’s weight was reduced if food is to be used as a reward during ensuing experiments. Hunger motivation is very effective in conditioning the pigeon, but the bird must be at approximately 80% of his ad lib weight.

The food and water was given only 5 gram to reduce the weight, until the bird reaches 80%. When the pigeon reaches 80% of his ad lib weight, feed him up to that weight every day. Feed as close as possible to the time you will be experimenting. On experimental days, the bird will be a few grams below 80% if he was fed up to that with the preceding day. During experimentation he will probably gain back to 80%. If it does not, bring it up to that rate in the end of the experimental session.

The ad libitum weight of the pigeon was calculated by taking an average of its weight of 3 days i.e., 15 to 17 November 2020.

When ad lib is determined, the pigeon’s weight must be reduced if food is to be used as a reward during in ensuing experiments. For calculating the experimental weight, 80% of the ad lib weight is been taken. The experimental weight of our pigeon was:

Experimental Weight = 80% of Adlib weight

= 80/100 x 272.3

= 217.84 grams

Reduction of Weight

The weight of the pigeon was planned to reduce so that it reaches its experimental weight 4-5 grams food was given one time a day for this purpose. The food quantity was strictly maintained to attain the experimental weight. Pigeon was weighed every day to keep in record its weight and to know that when it will reach its experimental weight.

Skinner box

At first day pigeon was introduced with Skinner box and the food was provided in the chamber. That day is called chamber adaptation day. Chamber adaptation was given to the pigeon when there was 10 grams difference between its actual weight and calculated experimental weight. Before putting pigeon into the Skinner box, box was prepared and checked that whether it is working properly or not. Duties were assigned to all group members. One member controlled the panel, other noted the reading of every trial, and one noted the time with stopwatch and one observed pigeon during her procedure. In adaptation day house light remained on during whole procedure. That room in which chamber adaptation was performed was dark. Food was provided in food container which was placed right below the magazine. After doing all these essential preparations pigeon was placed inside the chamber. Firstly the pigeon slept for 15 minutes then he awaked up and walked around the chamber and observed its environment. After few minutes it went towards the container and started feeding after completion of housing pigeon was placed back into the cage.

Weight Chart

Date	Weight	Diet
6-11-2020	273	Free food
7-11-2020	277	Free food
8-11-2020	275	Free food
9-11-2020	276	5g

10-11-2020	272	5g
11-11-2020	273	5g
12-11-2020	274	5g
13-11-2020	270	5g
14-11-2020	272	5g
15-11-2020	273	5g
16-11-2020	272	5g
17-11-2020	269	5g
18-11-2020	258	5g
19-11-2020	259	5g
20-11-2020	262	5g
21-11-2020	268	5g
22-11-2020	267	5g
23-11-2020	263	5g
24-11-2020	265	5g

Magazine Training

At first the pigeon was weighed and was approximately 269 grams. Then the Skinner box was made ready i.e. cleaned from inside, checking of lights, functionality of switchboard, filling of the food magazine etc. The team of experimenters distributed the responsibilities of duties which were to be simultaneously performed during the experiment. The following duties were distributed amongst the team members: operating switchboard, observing the pigeon's behavior, controlling time for food magazine and noting down the number of responses.

After distributing and understanding the above-mentioned duties, the initial trials of the magazine training were started. The house light was turned on and remained on during all the trials. The pigeon was put in the Skinner box. There were a total of 30 trials. The key light or magazine light was turned on for approximately 5 second after the pigeon started responding. There was a time lapse of 5-15 seconds between all the trials after the pigeon started responding. The pigeon did not move at all towards the food magazine during the initial trials. Gradually it started moving and turned its face toward the magazine. It then started pecking after approximately 15 minutes inside the Skinner box. From 15th trial onward the pigeon pecked on all the trials. There was a delayed pecking response which may be due to the element of fatigue. The pigeon did not take a longer time to peck in their respective trials. The trials were completed after approximately 15 minutes inside the Skinner box. It was noted carefully that the pigeon only eat 5 gram of food from the food magazine. After the end of the training the pigeon was again weighed and was to 269.5 gram, and then it was returned it to its cage and water was given. Similar procedure was repeated second day.

Behavioral Shaping and Continuous Reinforcement

Problem Statement

Behavioral shaping and continuous reinforcement

Introduction

Shaping is a form of conditioning that leads subjects to complete an operant behavior. This process is also known as “approximation conditioning.” Psychologists reinforce successive approximations in order to reach the targeted, operant behavior.

Operant Behavior

Shaping involves operant behaviors. This is a behavior that meets two requirements:

- The behavior is freely completed by the subject
- There is no triggering stimulus

Operant behaviors are sometimes created by operant conditioning.

Background

Shaping is a concept that many pet owners find hard to grasp. We're used to making animals do things by leading them or pushing them into the behavior we want—and it is hard to believe that there is another way. Common sense tells us that there is no possible way to get an animal to do something it has never done before, doing nothing yourself but reinforcing spontaneous movements.

The word "shaping" is scientific slang for building a particular behavior by using a series of small steps to achieve it. Shaping allows you to create behavior from scratch without physical control or corrections, but rather by drawing on your animal's natural ability to learn.

Even B.F. Skinner did not start out training animals by capturing and shaping spontaneously offered behavior. Initially, he taught his laboratory animals to press levers and accomplish other tasks by making small changes in the environment: raising the height of a bar in small increments until an animal had to reach higher, or increasing the "stiffness" of a button so a pigeon learned to peck harder. This method was called successive approximation.

In 1943, while waiting for a government grants to come through, Skinner and two of his graduate students decided to see if they could teach one of their experimental pigeons to bowl in a laboratory on the top floor of a building in Minnesota.

They started by putting the pigeon and a wooden ball in a box rigged with an automatic feeder, planning to trip the feeder when the pigeon swiped at the ball with its beak. But the pigeon did not swipe at the ball as they had hoped, and they grew tired of waiting. Skinner decided to reinforce any movement toward the ball, even just one look toward it. When the pigeon looked in that direction, he clicked the switch, opening the feeder briefly so the pigeon could get a bit of corn.

Skinner later wrote, "The result amazed us. In a few moments, the ball was caroming off the walls of the box as if the pigeon had been a champion squash player." Skinner had made a discovery that astonished even him: It was much easier to shape behavior by hand than by changing the environment.

Skinner's daughter, behavior analyst Julie Vargas, Ph.D., has told me, "His realization at that moment was that if you could do this, you could shape behavior anywhere, in any environment." You did not need to manipulate the task or build elaborate apparatus. You could just reinforce moves in the right direction.

Skinner named this newly discovered method *shaping*, to differentiate it from the mechanical process of successive approximation.

Methods of Shaping

There are five ways of shaping behavior.

Positive Reinforcement

It occurs when a desirable event or stimulus is given as an outcome of a behavior and the behavior improves. A positive reinforce is a stimulus event for which an individual will work in order to achieve it.

Negative Reinforcement

It occurs when an aversive event or when a stimulus is removed or prevented from happening and the rate of a behavior improves. A negative reinforce is a stimulus event for which an individual will work in order to terminate, to escape from, to postpone its occurrence.

Punishment

The creation of some unpleasant conditions to remove an undesirable behavior.

For example − A teenager comes home late and the parents take away the privilege of using the cell phone.

Extinction

The process of eradicating any type of reinforcement causing any undesirable behavior.

Schedules of Reinforcement

The schedules of reinforcement can be of five types – continuous, fixed interval, variable interval, fixed ratio, and variable ratio.

Continuous

A schedule of reinforcement in which every occurrence of the desired outcome is followed by the one who reinforces.

Fixed interval

Conduct of reinforcement with intervals but sufficient enough to make the expected behavior worth repeating.

Variable interval

Conduct of reinforcement with an average of n amount of time.

Fixed ratio

Oversight of reinforcement when rewards are spaced at uniform time intervals.

Variable ratio

Oversight of reinforcement when rewards are spaced at unpredictable time intervals.

Hypothesis

Behavior gets strengthened through successive approximation by using continuous reinforcement.

Method

- **Subject**

Subject: Pigeon

Sex: Male

Age: 2-3 years

- **Apparatus**

Skinner box, food, stop-watch, paper, pencil, weighing instrument and cage.

- **Procedure**

The process of behavior shaping of the pigeon was completed in three sessions. The pigeon was approximately nearest to the experimental weight. Magazine training was completed. The pigeon was fully prepared for the experiment. The experimenters were fully vigilant during the experiment and all responsibilities were assigned to the group members. The pigeon was taken out from his cage and was weighed before the experiment. The food was put in the food magazine of Skinner box. The pigeon was then placed in the Skinner box the house light remained off and only stimulus light remained on during the experiment. The experimenter started to note down pigeon's behavior and responses. The required response from the pigeon was his pecking on the Yellow light and food was provided as reinforcement (reward). When the pigeon came close to the light the food magazine was on to provide food by successive approximation. By successive approximation pigeon started to peck on the Yellow light, his numbers of pecks were noted. The time of peck was also noted by stop watch. After pecking food was provided through food magazine and the pigeon was allowed to eat for only five seconds. When pigeon started pecking on Yellow light and after food returned back for pecking, it was recorded as first response

Results

Table 1

Shaping day 1

265g

Tiles	Peck	VITI
1	1	3 minutes 56 sec
2	1	1 minutes 37 sec
3	1	59 sec
4	1	27 sec
5	1	9 sec
6	1	15 sec
7	1	12 sec
8	2	10 sec
9	1	28 sec
10	1	37 sec
11	1	40 sec
12	1	1 minute 8 sec
13	1	2 minutes 18 sec
14	3	26 sec

15	4	11 sec
16	3	26 sec
17	3	28 sec
18	3	15 sec
19	3	12 sec
20	3	14 sec
21	3	20 sec
22	1	45 minutes 30 sec
23	1	33 sec
24	1	20 sec
25	1	21 sec
26	2	7 sec
27	1	5 sec
28	1	10 sec
29	1	13 sec
30	1	10 minutes 24 sec

Note. VITI= Variable inter trial interval

Table 02

Shaping day 02

272 g

Trials	Peck	VITI
1	1	29sec
2	1	35sec
3	1	04sec
4	1	25sec
5	1	09 sec
6	1	09 sec
7	1	12 minutes 9 sec
8	1	5 minutes
9	1	11 sec
10	1	24 sec
11	1	28 sec
12	2	3 minutes 26 sec
13	2	16 sec
14	1	28 sec
15	1	16 sec

16	3	35 sec
17	1	39 sec
18	1	37 sec
19	1	3 minutes 7 sec
20	1	41 sec
21	1	1 minutes 10 sec
22	1	1 minutes 3 sec
23	2	9 sec
24	1	54 sec
25	1	18 sec
26	1	2 sec
27	1	1 minutes 1 sec
28	2	2 minutes 18 sec
29	1	20 minutes 3 sec
30	1	50 sec

Note. VITI= Variable inter trial interval

Table 3

Shaping day 3

269g

Trials	Peck	VITI
1	1	10 minutes 18 sec
2	1	17 sec
3	2	22 sec
4	2	37 sec
5	1	8 minutes 32 sec
6	1	22 minutes 32 sec
7	1	9 sec
8	2	22 minutes 39 sec
9	3	10 sec
10	2	23 sec
11	3	36 sec
12	3	12 sec
13	1	12 sec
14	2	16 sec
15	3	41 minutes 10 sec

16	1	34 sec
17	2	32 sec
18	3	1 minute 18 sec
19	1	1 minute 42 sec
20	1	40 sec
21	3	11 sec
22	4	8 sec
23	3	5 sec
24	2	12 sec
25	2	15 sec
26	1	22 sec
27	1	1 minute 2 sec
28	1	10 sec
29	2	5 sec
30	1	4 sec

Note. VITI= Variable inter trial interval

Table 4

Results showing averages of the no. of pecks

Trials	Peck	VITI (Sec)
10	1	51
20	2.5	38
30	1.3	345

Table 5

Results showing averages of the no. of pecks

Trials	Peck	VITI (Sec)
10	1	117.5
20	1.4	63.3
30	1.2	166

Table 6

Results showing averages of the no. of pecks

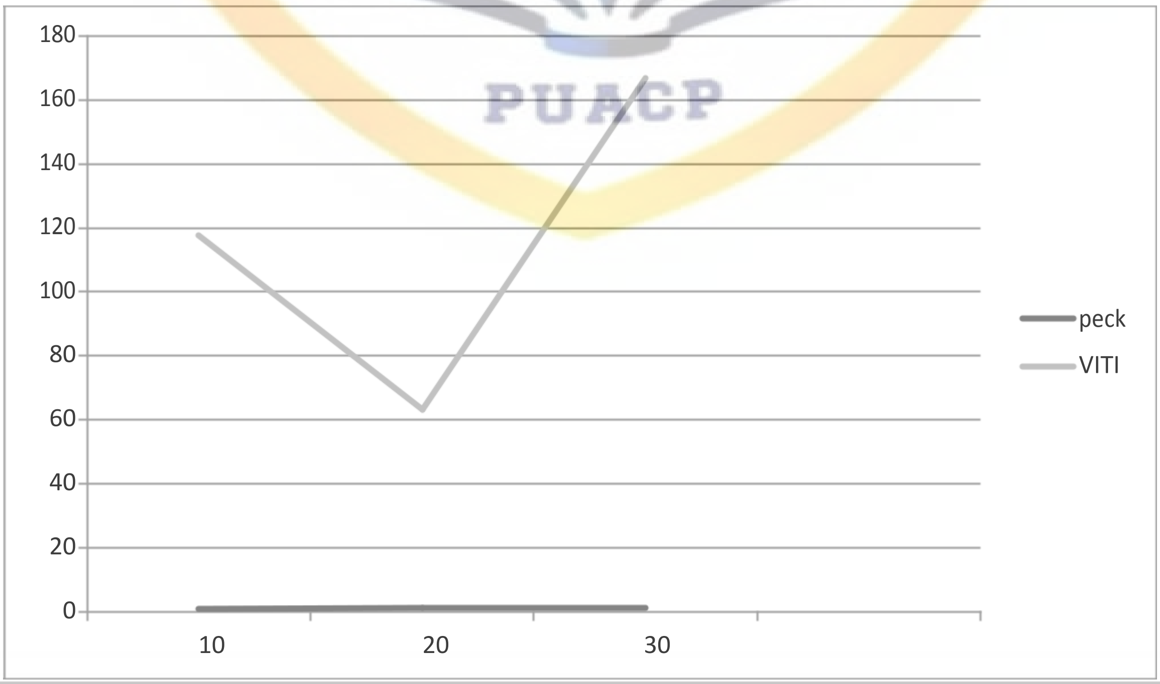
Trials	Peck	VITI (Sec)
10	1.6	395
20	2	285
30	2	15.2

Graph 1

Averages of the no. of pecks

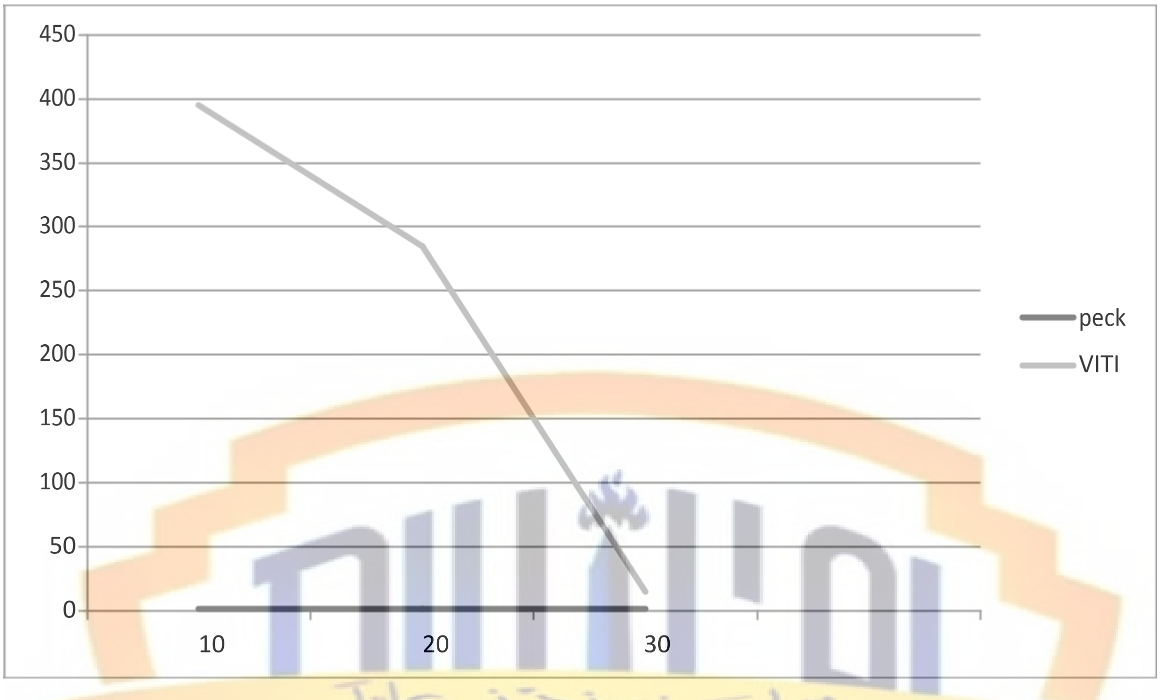
Graph 2

Averages of the no. of pecks



Graph 3

Averages of the no. of pecks



Discussion

The process of behavior shaping of the pigeon was completed in three sessions.

On the first day of shaping, the pigeon was very lazy and during the session pigeon was slept many times. Experimenters gave food to the pigeon on one or two pecks. In first trials pigeon VITI was low because pigeon had been hungry for twenty four hours. But after some trials pigeon VITI increase. Even in twenty second trial pigeon VITI was 45 minutes because the pigeon was asleep. May be it was the reason of fatigue. But after sleeping, the pigeon was stabbed again.

On the second day, pigeon pecked very easily and didn't take much time because the pigeon had a lot of shaping. But due to fatigue, the pigeon took a while for the last turn.

On the third day, the pigeon was shaped very well. But its VITI was increase in some trials because the pigeon was sleeping again and again.

Generalization

Problem Statement

If the pigeon learn to respond on a stimulus (peck) does she carry out the same response (pecking) on the stimulus having same characteristics like the first stimulus?

Introduction

In psychology, generalization is the tendency to respond in the same way to different but similar stimuli.

Stimulus Generalization Process is Conditioned

In the conditioning process, stimulus generalization is the tendency for the conditioned stimulus to evoke similar responses after the response has been conditioned. For example, if a child has been conditioned to fear a stuffed white rabbit, it will exhibit a fear of objects similar to the conditioned stimulus such as a white toy rat.

One famous psychology experiment perfectly illustrated how stimulus generalization works. In the classic Little Albert experiment, researchers John B. Watson and Rosalie Rayner conditioned a little boy to fear a white rat.

The researchers observed that the boy experienced stimulus generalization by showing fear in response to similar stimuli including a dog, a rabbit, a fur coat, a white Santa Claus beard, and even Watson's own hair.

Instead of distinguishing between the fear object and similar stimuli, the little boy became fearful of objects that were similar in appearance to the white rat. (Though it should be noted, this experiment has been the subject of much debate and controversy in recent years)

Importance

It is important to understand how stimulus generalization can influence responses to the conditioned stimulus. Once a person or animal has been trained to respond to a stimulus, very similar stimuli may produce the same response as well. Sometimes this can be problematic, particularly in cases where the individual needs to be able to distinguish between stimuli and respond only to a very specific stimulus.

The importance of the generalization of skills is often overlooked. When someone learns new skills and behaviors, the expectation is that he or she will automatically generalize them. However, educators and analysts who work with children and adults with learning deficits and other challenges must not only teach new skills, but also employ additional strategies to increase the likelihood that their students will be able to generalize those skills.

Hypothesis

Stimulus SI (Yellow light) and S2 (Red light) are rewarded equally to strengthen the behavior.

Method

- **Subject**

Subject:	Pigeon
Sex:	Male
Age:	2-3 years

- **Apparatus**

Skinner box, food, stop-watch, paper, pencil, weighing instrument and cage.

- **Procedure**

The main purpose of this experiment was to generalize the pigeon behavior that if the pigeon learn to respond on a stimulus (Yellow light) does he carry out the same response (Pecking) on the stimulus having same characteristics (Red light) like first stimulus and then he will get the food on both lights Yellow and red light. Firstly the Skinner box was prepared or checked by the experimenter that all lights are working or not. Then the magazine was filled with food. After that the pigeon was weighted before the experiment. He was on experimental weight and ready for the experiment. Then he was put in the Skinner box. The experiment was conducted in a dark room. There was complete silence in the room throughout the experiment. The starting time of the experiment was noted. As experiment was conducted in group, duties were assigned to all the group members. Total 30 trials were made. The pigeon's behavior was reinforced or generalizes on the both Red (S1) and Yellow (S2) lights. Whenever, the pigeon pecked on the light either Yellow and red, he was reinforced by food through magazine for 5 seconds. His behavior was fully observed throughout the experiment.

The general trend was seen in pigeon's behavior was that he started pecking on Yellow light at the start of the experiment actively and sudden change in light color (red) cause a little passiveness in pigeon but after the number of trials the pigeon equally respond on both lights (pecking). Whenever he pecked on the light, he was reinforced immediately. The pigeon learned that he will get food by pecking on both lights. So, he pecked on both yellow and red lights and was reinforced. In this way 30 trials were made.

Results

Table 1

Generalization

272 g

Trials	Stimulus key	Pecks	Reward	VITI
1	YL	1	5 sec	5 minutes 27 sec
2	YL	1	5 sec	8 minutes 25 sec
3	YL	1	5 sec	19 minutes
4	YL	3	5 sec	55 sec
5	YL	2	5 sec	10 sec
6	YL	4	5 sec	20 sec
7	YL	3	5 sec	11 sec
8	GL	3	5 sec	49 sec
9	GL	3	5 sec	13 minutes
10	YL	2	5 sec	60 sec
11	GL	3	5 sec	26 minutes 49 sec
12	YL	1	5 sec	41 minutes 39

				sec
13	GL	3	5 sec	40 sec
14	GL	3	5 sec	1 minuit 45 sec
15	YL	1	5 sec	14 minutes 39 sec
16	GL	3	5 sec	23 sec
17	YL	2	5 sec	51 sec
18	GL	3	5 sec	22 sec
19	YL	3	5 sec	30 sec
20	GL	3	5 sec	21 sec
21	YL	3	5 sec	35 sec
22	GL	3	5 sec	8 sec
23	YL	3	5 sec	43 sec
24	GL	2	5 sec	15 sec
25	YL	3	5 sec	11 sec
26	GL	3	5 sec	23 sec
27	GL	3	5 sec	10 sec
28	GL	3	5 sec	13 sec

29	GL	2	5 sec	56 sec
30	GL	3	5 sec	21 sec

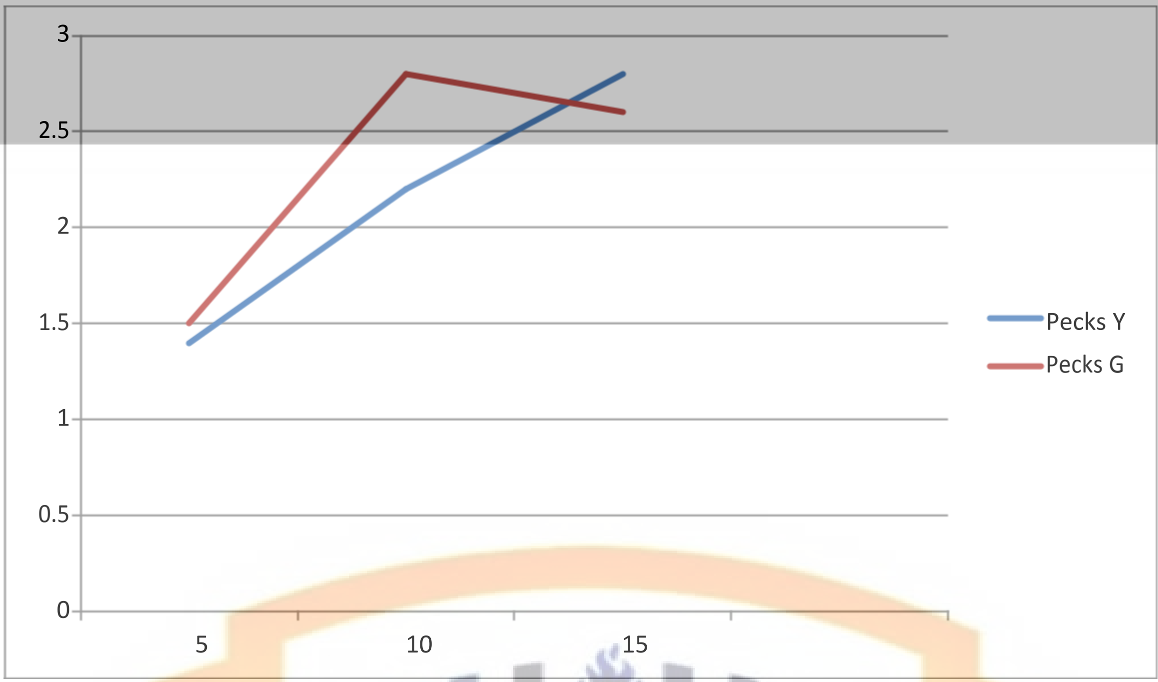
Table 2

Results showing averages of the No. of pecks on yellow and red light

No of Trials	No of Pecks		VITI (Sec)	
	Yellow light	**Green light**	**Yellow light**	**Green light**
5	1.4	1.5	203	264
10	2.2	2.8	394.9	8.9
15	2.8	2.6	104.9	12.5

Graph

Showing averages of pecks on yellow and red light.



Discussion

After shaping, experimenters generalized the pigeon. Due to the lockdown and pandemic condition experimenters did generalization in only one day. Experimenters did 15 trials on yellow light and 15 trials on green light. In some trials VITI was increase and in some trials VITI was decrease. However, at the end of the session the pigeon generalized very well.

Stimulus Discrimination

Problem Statement

To find out the either organisms (pigeon) is able to distinguish between two similar stimuli. Or respond differently to two or more similar stimuli.

Introduction

In psychology, discrimination is the ability to perceive and respond to differences among stimuli. It is considered a more advanced form of learning than generalization (*q.v.*), the ability to perceive similarities, although animals can be trained to discriminate as well as to generalize.

Different consequences may follow the same behavior in different situations. When we respond differently in those different situations, we have formed a discrimination between the situations. For instance, when you tell a ribald tale to friends at a party, but refrain from doing so at a church gathering, this is an example of discrimination. A past history of positive reinforcement in the first case and a past history of positive punishment in the second could clearly be responsible for this illustration of discriminative responding.

Failure to have discriminated between these different situations would represent a case of inappropriate stimulus generalization.

Discriminative Stimulus

"Any stimulus present when an operant is reinforced acquires control in the sense that the rate will be higher when it is present. Such a stimulus does not act as a goad; it does not elicit the response in the sense of forcing it to occur. It is simply an essential aspect of the occasion upon which a response is made and reinforced. The difference is made clear by calling it a discriminative stimulus..."(Skinner, 1969).

Discrimination in Classical Conditioning

In classical conditioning, discrimination is the ability to differentiate between a conditioned stimulus and other stimuli that have not been paired with an unconditioned stimulus. For example, if a bell tone were the conditioned stimulus, discrimination would involve being able to tell the difference between the bell sound and other similar sounds.

The classical conditioning works like this: A previously neutral stimulus, such as a sound, is paired with an unconditioned stimulus (UCS). The unconditioned stimulus represents something that naturally and automatically triggers a response. For example, the smell of food is an unconditioned stimulus, while salivating to the smell is an unconditioned response.

After an association has been formed between the previously neutral stimulus, now known as the conditioned stimulus (CS), and the unconditioned response, the CS can evoke the same response, now known as the conditioned response, even when the UCS is not present.

In Ivan Pavlov's classic experiments, the sound of a tone (a neutral stimulus that became a conditioned stimulus) was repeatedly paired with the presentation of food (unconditioned stimulus), which naturally and automatically led to a salivary response (unconditioned response).

Eventually, the dogs would salivate in response to the sound of the tone alone (a conditioned response to a conditioned stimulus). Now, imagine that Pavlov introduced a different sound to the experiment. Instead of presenting the sound of the tone, let's imagine that he sounded a trumpet. What would happen?

If the dogs did not drool in response to the trumpet noise, it means that they are able to discriminate between the sound of the tone and the similar stimulus. Not just any noise will produce a conditioned response. Because of stimulus discrimination, only a very particular sound will lead to a conditioned response.

In one well-known experiment on classical conditioning, researchers paired the taste of meat (unconditioned stimulus) with the sight of a circle (conditioned stimulus), and dogs learned to salivate in response to the presentation of a circle. The researchers found, however, that the dogs would also salivate when they saw an ellipse, an oval shape.

Over time, as the dogs experienced more and more trials where they did not experience the taste of meat upon seeing the ellipse, they eventually became able to discriminate between the two similar stimuli. They would salivate in response to the circle, but not when they saw the ellipse.

Discrimination in Operant Conditioning

In operant conditioning, discrimination refers to responding only to the discriminative stimuli and not to similar stimuli. For example, imagine that you have trained your dog to jump in the air whenever you say the command, "Jump!" In this instance, discrimination refers to your dog's ability to distinguish between the command for jumping and similar commands such as sit, stay, or speak.

Importance

Discrimination is a term used in both classical and operant conditioning. It involves the ability to distinguish between one stimulus and similar stimuli. In both cases, it means responding only to certain stimuli, and not responding to those that are similar.

Hypothesis

Stimulus SI (Red key light) and stimulus S2 (Yellow light) are not potential equally to strengthen the behavior.

Method

- **Subject**

Subject: Pigeon

Sex: Male

Age: 2-3 years

- **Apparatus**

Skinner box, food, stop-watch, paper, pencil, weighing instrument and cage.

- **Procedure**

Experiment was performed in an experimental lab. It was quiet peaceful and comfortable place. The pigeon’s weight was 230 gram on that day. There was complete darkness and silence maintained in the experimental lab. The chamber was made ready. Responsibilities were assigned to all the experimenters e.g. panel control, time check, record the trial and observe the behavior of the organism. In this stimulus discrimination experiment, experimenters wanted the pigeon to discriminate different stimuli that were red as rewarding and yellow light as non-rewarding stimuli. When the pigeon pecked on Yellow light, he would not be reinforced but when he pecked on the red light he would be reinforced. After finalizing all the steps of the experiment, the experiment was started the pigeon was brought to the chamber.

The stimuli were given for 5 seconds randomly. Firstly we mostly made red light on so the pigeon may ate food more, then we introduce a Yellow light (non-rewarding). The magazine light was switched on for 5 seconds after pecking on red light. In the first 15 trials, he constantly pecked on both stimuli lights because in the earlier experiment, he learned that pecking on different stimuli lights was rewarding. But on later trials his pecking responses on Yellow light decreased as he was not reinforced when he pecked the yellow light.

In shot at the end of the experiment, the pigeon successfully discriminated which stimulus was rewarding for him and which was not.

Results

Table 1

Discrimination

270 g

Trials	Light	Pecks	VITI	Reinforcement
1	YL	1	3 minutes 48 sec	5 sec
2	YL	1	37 minutes 37 sec	5 sec
3	RL	3	27 sec	-
4	YL	1	1 minutes 21 sec	5 sec
5	RL	2	2 minutes 09 sec	-

6	YL	1	21 sec	5 sec
7	RL	2	3 sec	-
8	YL	3	3 sec	5 sec
9	RL	1	51 sec	-
10	YL	1	35 minutes 28 sec	5 sec
11	YL	3	19 sec	5 sec
12	RL	2	39sec	-
13	YL	1	6 minutes 44 sec	5 sec
14	RL	2	38 sec	-
15	YL	3	45 sec	5 sec
16	RL	2	27 sec	-
17	RL	1	37 sec	-
18	YL	3	15 sec	5 sec
19	YL	3	5 sec	5 sec
20	RL	2	3 sec	-
21	YL	3	4 sec	5 sec
22	RL	1	5 sec	-

23	YL	1	7 sec	-
24	YL	2	11 sec	5 sec
25	RL	1	6 sec	-
26	YL	1	5 minutes 27 sec	5 sec
27	RL	0	0	5 sec
28	RL	0	0	5 sec
29	RL	0	0	-
30	RL	0	0	-

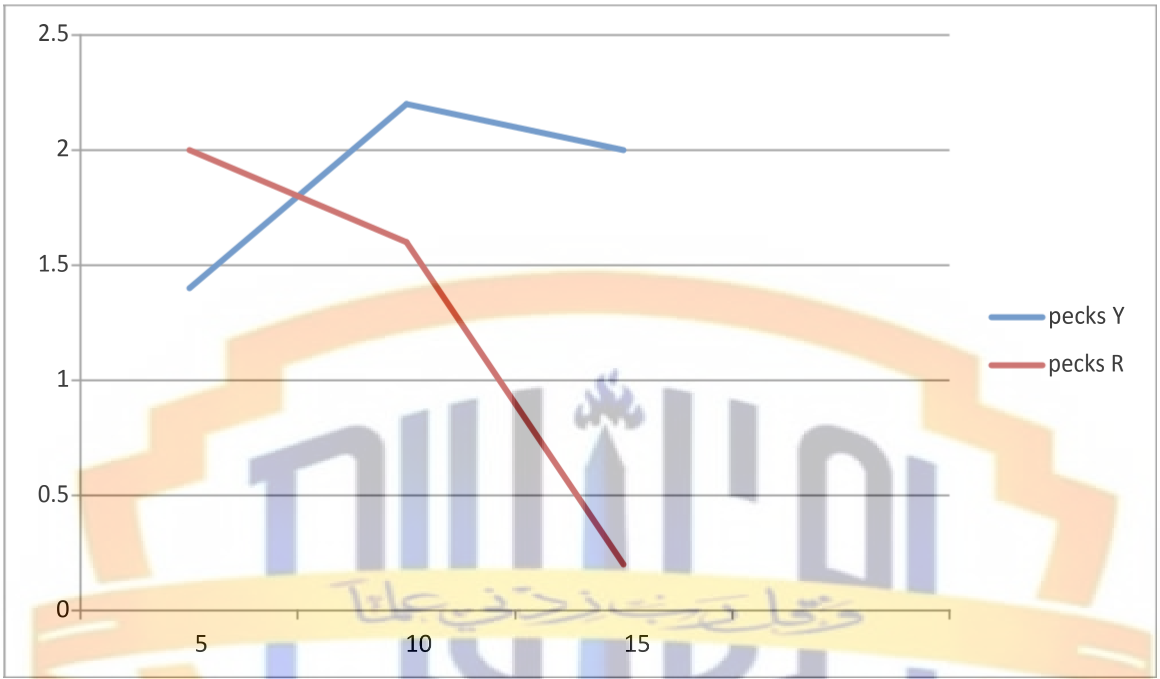
Table 2

Results showing averages of the no. of pecks

No of Trials	No of Pecks		VITI (Seconds)	
	Yellow light	**Red light**	**Yellow light**	**Red light**
5	1.4	2	259	50
5	2.2	1.6	522.2	22
5	2	0.2	69.2	1.2

Graph

Graph showing averages of the no. of pecks



Discussion

After generalization, experimenters discriminate the pigeon. Due to the lockdown and pandemic condition experimenters did discrimination in only one day. Experimenters did 15 trials on yellow light and 15 trials on red light. In some trials VITI was increase and in some trials VITI was decrease. Experimenters reinforced the pigeon for five seconds. However, at the end of the session the pigeon discriminate very well.

Extinction

Problem Statement

The behavior is likely to weaken after removal of reinforcement?

Introduction

Extinction in psychology refers to the fading and disappearance of behavior that was previously learned by association with another event.

In psychology, extinction refers to the gradual weakening of a conditioned response those results in the behavior decreasing or disappearing. In other words, the conditioned behavior eventually stops.

Extinction in Classical Conditioning

Classical conditioning occurs when an association is formed between a biologically significant natural stimulus and a neutral stimulus to cause an involuntary response.

The natural stimulus is an unconditioned stimulus (UCS) because it doesn't require any conditioning to cause the reaction. The neutral stimulus becomes a conditioned stimulus (CS), and the involuntary response becomes a conditioned response (CR).

The case of Pavlov's dogs is the most famous example. Ivan Pavlov rang a bell every time he fed his dogs. Eventually, they began to salivate every time they heard the sound, whether or not they received food. The natural salivation became a conditioned response.

In classical conditioning, extinction occurs when the conditioned stimulus is applied repeatedly without being paired with the unconditioned stimulus. Over time, the learned behavior occurs less often and eventually stops altogether, and conditioned stimulus returns to neural.

Extinction in Operant Conditioning

Operant conditioning refers to associating a natural stimulus (unconditioned stimulus) with reinforcement or punishment (conditioned stimulus) to change a voluntary behavior (conditioned response).

Operant extinction refers to the weakening and eventual stop of the voluntary, conditioned response.

For example, a child associates the sound of a microwave with her favorite snack, and she rushes into the kitchen. But after dad uses the microwave several times without making the snack, she gradually stops.

Factors That May Influence Extinction

A number of factors can influence how resistant a behavior is to extinction. The strength of the original conditioning can play an important role. The longer the conditioning has taken place and the magnitude of the conditioned response may make the response more resistant to extinction.

Behaviors that are very well established may become almost impervious to extinction and may continue to be displayed even after the reinforcement has been removed altogether. Some research has suggested that habituation may play a role in extinction as well. For example, repeated exposure to a conditioned stimulus may eventually lead you to become used to it, or habituated.

Because you have become habituated to the conditioned stimulus, you are more likely to ignore it and it's less likely to elicit a response, eventually leading to the extinction of the conditioned behavior.

Personality factors might also play a role in extinction. One study found that children who were more anxious were slower to habituate to a sound. As a result, their fear response to the sound was slower to become extinct than non-anxious children.

Extinction doesn't erase Previous Learning

Because of the possibility of spontaneous recovery and dependance on context, psychologists now believe that extinction is not an unlearning process. Rather, it is a form of *new learning*, called extinction learning

Instead of erasing previous learning, the individual learns a new association between the conditioned stimulus and the lack of unconditioned stimulus

Hypothesis

Absence of reinforcement will gradually omit the pecking response.

Method

- **Subject**

Subject:	Pigeon
Sex:	Male
Age:	2-3 years

- **Apparatus**

Skinner box, food, stop-watch, paper, pencil, weighing instrument and cage.

- **Procedure**

Firstly, the Skinner box was prepared for the experiment. The panel was checked whether it is operating or not and the magazines was filled with food. The house light was off during the whole experiment while the Yellow light was on. The pigeon was weighted and he was below from experimental weight. The process of conducting experiment on extinction involves omitting the pigeon responses. Precautionary measures were taken, in previous experiments food was provided to the pigeon in the Skinner box but not in the cage while in extinction the process is totally changed because in extinction food was provided to the pigeon in cage not in Skinner box.

Before starting the experiment, functioning of chamber/Skinner box was properly checked. It was made sure that the magazine was empty and there was no food in it. All the group members were assigned different responsibilities for example to control the panel, to record timing, to count number of pecks etc. Then the pigeon was weighted and its weight was noted. The pigeon was brought to the experimental room and was put in the Skinner box. Total thirty trials were given to the pigeon. Whenever, the pigeon pecked on the Yellow light, he was not reinforced with food. He pecked the light for so many times but no reinforcement was provided. At the end of the experiment, number of pecks/responses was reduced to zero. After finishing experiment, pigeon was provided with 7 grams of food in his cage.

Results

Table 1

Extinction

269g

Trials	Light	Pecks	VITI	Reinforcement
1	Chamber Light	2	2 minutes	-
2	Chamber Light	2	45 sec	-
3	Chamber Light	1	1 minute 35 sec	-
4	Chamber Light	3	15 sec	-
5	Chamber Light	1	20 sec	-
6	Chamber Light	2	13 sec	-
7	Chamber Light	2	30 sec	-
8	Chamber Light	2	27 sec	-
9	Chamber Light	3	10 sec	-
10	Chamber Light	3	15 sec	-
11	Chamber Light	1	3 minutes 23 sec	-
12	Chamber Light	3	15 sec	-
13	Chamber Light	2	10 sec	-
14	Chamber Light	2	17 sec	-
15	Chamber Light	3	11 sec	-
16	Chamber Light	1	35 sec	-

17	Chamber Light	2	20 sec	-
18	Chamber Light	1	35sec	-
19	Chamber Light	1	1 minute	-
21	Chamber Light	2	35 sec	-
22	Chamber Light	1	45 sec	-
23	Chamber Light	1	1 minutes 23 sec	-
24	Chamber Light	1	1 minutes 45 sec	-
25	Chamber Light	0	0	-
26	Chamber Light	0	0	-
27	Chamber Light	0	0	-
28	Chamber Light	0	0	-
29	Chamber Light	0	0	-
30	Chamber Light	0	0	-
31	Chamber Light	0	0	-

Table 2

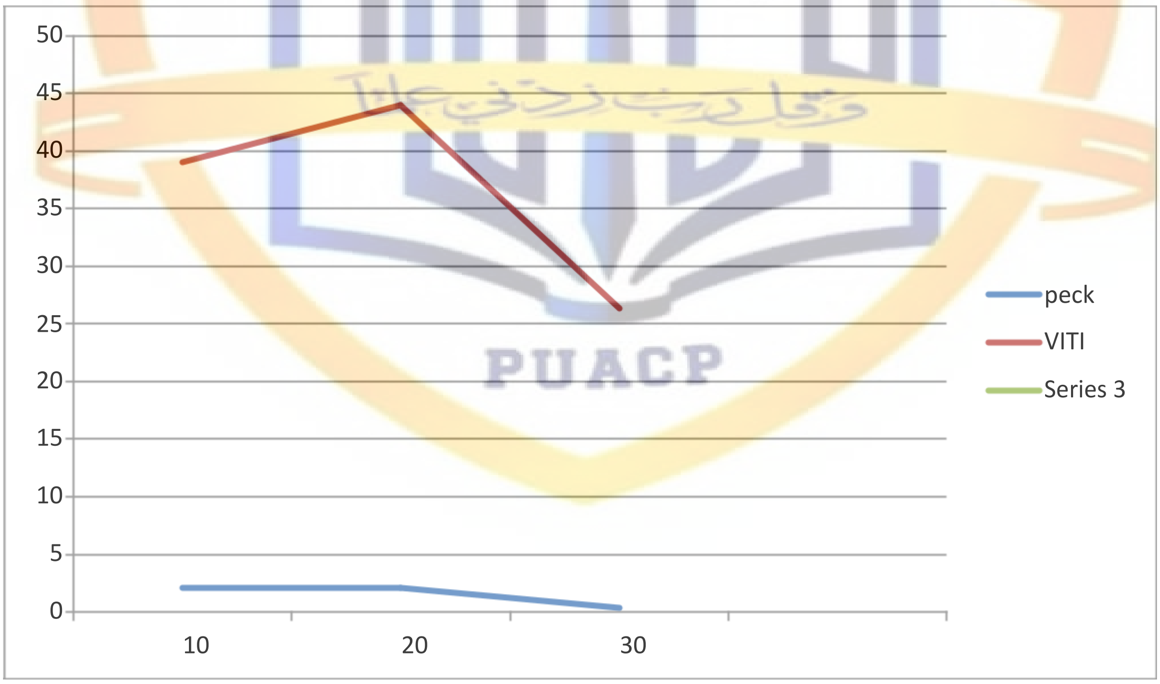
Results showing averages of pecks

Trials	Peck	VITI (Sec)

10	2.1	39
20	2.1	44.1
30	0.4	26.3

Graph

Graphical representation of the no. of pecks



Discussion

After discrimination, experimenters did extinction of the pigeon. Due to the lockdown and pandemic condition experimenters did discrimination in only one day. Experimenters use chamber light for extinction. In two or three pecks of the pigeon experimenters did not reinforced the pigeon. Even a time came when pigeon stopped pecking. However, at the end of the session, number of pecks was reduced to zero. The experimenters did the extinction with pigeon very well.

Spontaneous Recovery

Problem Statement

To find out either reconditioning takes less time than conditioning.

Introduction

Spontaneous recovery refers to the sudden reappearance of a previously extinct conditioned response after the unconditioned stimulus has been removed for some time.

Spontaneous recovery is a phenomenon that involves suddenly displaying a behavior that was thought to be extinct. This can apply to responses that have been formed through both classical and operant conditioning.

Spontaneous recovery can be defined as the reappearance of the conditioned response after a rest period or period of lessened response. If the conditioned stimulus and unconditioned stimulus are no longer associated, extinction will occur very rapidly after a spontaneous recovery.

Spontaneous recovery can occur after two types of conditioning have taken place.

Spontaneous Recovery in Classical Conditioning

In classical conditioning, also known as pavlovian conditioning, a previously neutral stimulus (NS) becomes conditioned when it is repeatedly paired with the unconditioned stimulus (US). This now conditioned stimulus (CS) can produce its own conditioned response (CR), which is usually very similar to the unconditioned response (UR).

However, some conditioned responses are vulnerable to extinction. If the conditioned stimulus continues to appear in the absence of the unconditioned stimulus, the conditioned response becomes weaker and weaker until it disappears, which is called the extinction procedure.

A famous example is Pavlov’s dogs. Ivan Pavlov, who coined the term pavlovian conditioning, experimented by sounding a bell repeatedly when he fed the dogs. Over time, they learned to associate the sound with food and salivated when they heard the sound. Then Pavlov started ringing the bell without giving food. Eventually, the dogs stopped salivating to the sound of the bell.

However, Pavlov noticed that even after a substantial amount of time had passed, the conditioned response would easily recover if the neutral stimulus and the unconditioned stimulus were paired again. Also though the dogs stopped salivating to the sound of the bell, their salivation recovered spontaneously after a “rest period.”

Another example is that a child gets excited every day when they hear the ice cream truck music because their mother always buys them ice cream. When their mother stops buying, the child gradually learns to not associate the ice cream truck music with eating ice cream. After the truck stops coming for a few days and then returns, the child gets excited again when they hear the truck music.

Spontaneous Recovery in Operant Conditioning

A trainer teaches a dog to sit by associating the command “Sit” with food. So the dog learns to sit whenever the trainer says the word. But after the trainer stops giving it food, it gradually stops responding to the command. Days later, the trainer tries again, and the dog sits.

Here is another example. A child runs to the door to greet Dad because he always brings home a new toy. After Dad stops bringing home toys, the child stops running to the door to greet him. After a few days, the child suddenly starts greeting their Dad at the door again.

Process of Spontaneous Recovery

The fact that conditioned response can spontaneously recover suggests that extinction doesn't erase the learned association. Instead, extinction inhibits the conditioned response. It appears that extinction forms new learning separate from the original conditioned learning.

This new learning "extinguish" the conditioned response by inhibiting its expression instead of erasing or unlearning it.

Since the initial conditioned response never disappears, it eventually returns. Studies show that, with sufficient time, spontaneous recovery of the fear response occurs 100% in situations such as fear conditioning.

Since new learning does not replace old ones, spontaneous recovery does not replace the extinction learning either. It simply exists in the presence of extinction learning.

The extinction process creates a new learning memory. During spontaneous recovery, the reactivated memory from extinction competes with the reactivated memory from initial conditioning but fails.

Importance

But if a conditioned response becomes extinguished, does it really disappear altogether? If Watson and Rayner had next given the boy a brief rest period before reintroducing the rat, Little Albert might have exhibited a spontaneous recovery of the fear response. Why is spontaneous recovery so significant?

After a conditioned response has been extinguished, spontaneous recovery may gradually increase as time passes. However, the returned response will generally not be the same strength as the original response unless additional conditioned takes place.

Numerous cycles of extinction followed by recovery usually result in progressively weaker responses. Spontaneous recovery may continue to take place, but the response will be less intense.

Hypothesis

"Reconditioning takes less time than conditioning"

Method

- **Subject**

Subject:	Pigeon
Sex:	Male
Age:	2-3 years

- **Apparatus**

Skinner box, food, stop-watch, paper, pencil, weighing instrument and cage.

- **Procedure**

The experiment was performed in one session. Magazine was filled with food. It was sure that no grain of food was on the floor of Skinner box. It was also checked that the hatchet was firmly attached but swing freely to admit or prevent access to the food. Then all lights including house light was turned off. All the group members were assigned different responsibilities for example to control the panel, to record timing, to count number of pecks etc. Then pigeon was weighted, brought to the experimental room and was put in the Skinner box. A table was prepared before experiment in which 15 trials of red light and 15 trials of Yellow light were randomly distributed.

Starting time of the experiment was noted. There was not fixed or variable interval. Whenever the pigeon pecks on the light, reinforcement was provided. Total 30 trials were taken for completing the experiment. Pecking responses were noted during every trial. Time was noted using stopwatch, which the pigeon took after taking reinforcement for very next peck on the light. Behavioral observation was noted during the whole experiment by one of the group member. During first 10 trials, the pigeon's response was a little late. The pigeon started to peck on 5th second and onward on the light after taking the reinforcement but after first 10 trials, the pigeon learnt how to respond for getting food. The pigeon was reinforced after very trial for 4 seconds. All the 30 trials were taken in the same way and response were noted. After completing 30 trials, the pigeon was brought from the chamber box out carefully and weighted again for seeing how much the pigeon has eaten. After weighing, the pigeon was kept into the cage. No food was provided in the cage.

Results

Table 1

Spontaneous recovery

245g

Trials	Peck	VITI	Reinforcement
1	1	1 minute	-
2	2	1 minute	-
3	2	1 minute	-
4	1	1 minute	-
5	1	1 minute	-
6	2	45sec	-
7	1	1 minute 50 sec	-
8	4	38 sec	-
9	1	3 minutes	-
10	1	50 sec	-
11	2	8 sec	-
12	1	32 sec	-
13	1	20 sec	-
14	1	1 minutes 2 sec	-
15	2	2 minutes 40 sec	-
16	0	0	-
17	0	0	-

18	0	0	-
19	0	0	-
20	0	0	-
21	0	0	-
22	0	0	-
23	0	0	-
24	0	0	-
25	0	0	-
26	0	0	-
27	0	0	-
28	0	0	-
29	0	0	-
30	0	0	-

Table 2

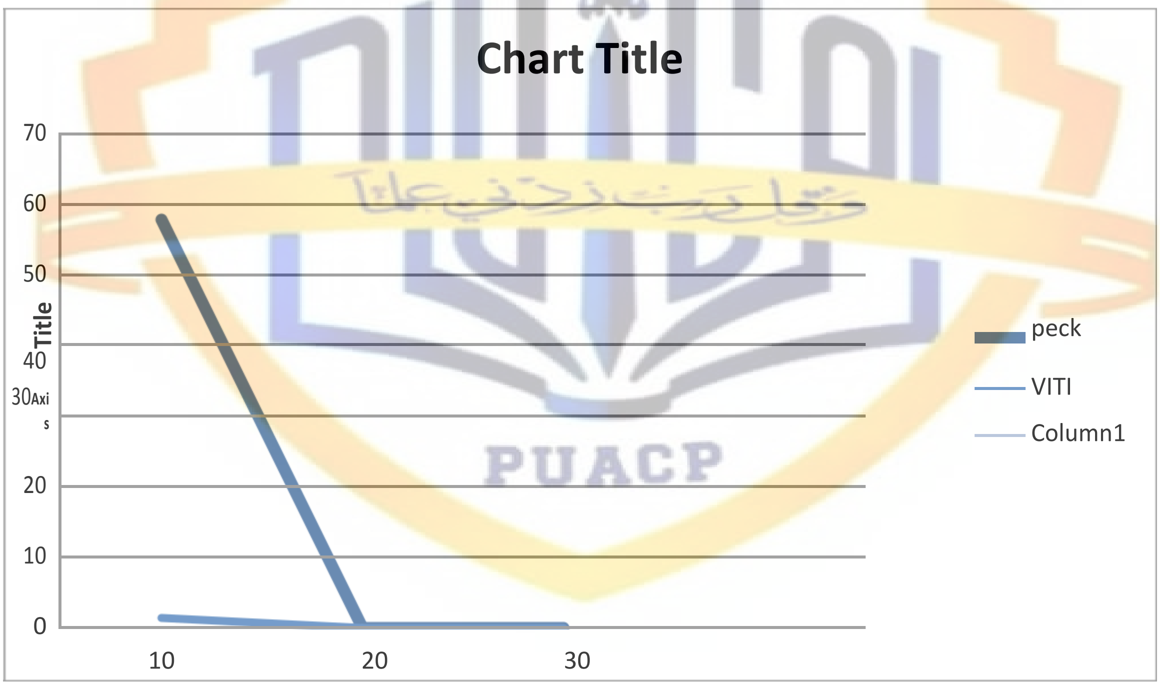
Averages of the No. of pecks on key light for Spontaneous Recovery

Trials	**Peck**	**VITI (Sec)**
10	57.8	1-3

20	0	0
30	0	0

Graph

Showing averages of the No. of pecks on key light



Discussion

After extinction, experimenters did spontaneous recovery of the pigeon on pecking and reinforcement. Due to pandemic situation experimenters did this on only one day. The pigeon was spontaneously recovered on learned behavior. During spontaneous recovery, it resumed its pecking within the first few trials.

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